A Data Driven Analysis of Global Shipping Emissions

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Abstract

The global shipping industry represents a substantial source of anthropogenic greenhouse gas (GHG) emissions, with projected trajectories indicating sustained growth in the absence of effective mitigation frameworks and policies. Establishing an equitable and effective framework to distribute responsibility for international maritime emissions among nations remains an unresolved challenge. Furthermore, developing a reliable methodology to accurately quantify these emissions using empirical data is essential for informing allocation decisions. Investigating this issue, Selin et al. (2021) utilized 2015 AIS data and concluded that allocating emissions based on the owner country best aligns with criteria for effectiveness and equity. Building upon their work, this paper introduces an open-data methodological framework designed to reconstruct their analysis, relying strictly on publicly accessible sources. Through the demonstration of this pipeline on two ships, COSCO ITALY and RCC AMERICA, we characterize the inherent constraints of such substitutions and specify how these challenges may be addressed to scale the procedure for a national allocation study. Obtaining ship specifications presents a greater challenge. The Equasis database does not allow bulk download, and no alternative open-source platform provides the necessary parameters to reliably estimate main engine power demand, a critical variable for CO_2 estimation. This uncertainty showed a 1.90x difference in total emissions for COSCO ITALY. Thus, the limiting factor for this methodology is not the model but a single register field, and any extension of this work should acquire installed power and reference speed first.

1. Introduction

International maritime shipping serves as a key element of global trade, transporting 70-80% of global trade volume (Zhao et al., 2025; Wang et al., 2026). Yet, it comes with a considerable climate cost. The Fourth IMO GHG Study estimated that greenhouse gas emissions from total shipping increased from 977 million tonnes in 2012 to 1,076 million tonnes in 2018, raising the sector's share of global anthropogenic emissions from 2.76% to 2.89% (International Maritime Organization [IMO], 2020). Similarly, in 2022 international shipping accounted for approximately 2% of global energy-related CO_2 emissions (International Energy Agency [IEA], 2023). Furthermore, the International Maritime Organization (IMO, 2020) estimates an increase in emissions between 10-30% by 2050, if there are no new regulations to reduce shipping impact on emissions, defined as the business-as-usual scenario.

Despite this contribution, the sector has historically occupied a governance vacuum. Heitmann and Khalilian (2011) explained that "emissions from international shipping are not assigned to national emissions inventories. Therefore, they remain unaffected by the Kyoto regulations" (p. 682). The Kyoto Protocol instead delegated these emissions to the IMO (Selin et al., 2021). Similarly, while national targets under the UNFCCC dictate mitigation policies for all other sectors, a complication arises for shipping concerning the allocation of responsibility for emissions released in international waters (Bows-Larkin, 2015). Despite many available technological and operational options, the industry's complex organizational nature is a major barrier to change. Experts have emphasized that "addressing GHG emissions from international shipping exceeds the capacities of one country or a small group of countries" (Hackmann, 2012, p. 99), and it requires a policy that considers the complexity of this activity.

Part of that complexity stems from the structural characteristics of international maritime shipping: ships are registered in one nation, owned in another, operated in a third, and traverse international waters, complicating standard national accounting (Oberthür, 2003). Lister et al. (2015) analyzed how this and other complexities in transnational shipping have caused lagging environmental regulation. Consequently, determining the most appropriate framework for distributing responsibility for international shipping emissions across countries remains a key unresolved challenge. Despite existing energy efficiency improvement plans for the global fleet, no single allocation option has been found to satisfy environmental, legal, and burden-sharing criteria simultaneously. This limitation demonstrates both the infeasibility of compulsory responsibility allocation under current frameworks and the necessity of a new mechanism design (Wang et al., 2021).

In April 2018, the IMO adopted its Initial Strategy, targeting a minimum 50% GHG reduction by 2050 relative to 2008 while pursuing complete phase-out (IMO, 2018). This strategy faced immediate criticism. Selin et al. (2021) highlighted institutional tension, noting the UNFCCC's explicit GHG focus versus the IMO's multifaceted agenda. Bullock et al. (2022) concluded Paris-compliant budgets required a 34% cut by 2030 and zero emissions before 2050, warning delays would demand untenable decarbonization rates. Bach and Hansen (2023) blamed insufficient policy on weak regulatory capacity, mandate uncertainty, and political disagreement, while Monios (2023) argued market-based mechanisms normalize pollution and displace direct regulation. Policy subsequently progressed. In July 2023, Resolution MEPC.377(80) replaced the Initial Strategy with a commitment to net-zero GHG emissions by or around 2050, adding indicative checkpoints for 2030 (20–30% reduction) and 2040 (70–80% reduction) relative to 2008 (IMO, 2023). Bilgili and Ölçer (2024) viewed this revision as more robust, despite persistent challenges like limited default emission factors.

In 1996, the UNFCCC's Subsidiary Body for Scientific and Technological Advice (SBSTA) outlined key national allocation options, such as national emissions, fuel sales, vessel registration or operator nationality, journey endpoints, cargo origin, and national territory, that remain central to contemporary debates (SBSTA, 1996). While foundational for research, their policy implementation remains limited. Equity, manifested through the principle of common but differentiated responsibilities and respective capabilities, seeks to balance mitigation costs using historical responsibility, capability, and vulnerability (Zenghelis, 2017). However, burden-sharing negotiations often push nations toward minimal commitments (Averchenkova et al., 2014). Given that allocation models produce similar results, Averchenkova et al. (2014) recommended prioritizing collaborative frameworks and development rights.

Applying these equity principles practically reveals notable gaps in national commitments. Robiou du Pont et al. (2017) modeled cost-optimal mitigation scenarios across five equity approaches, discovering that China's nationally determined contribution was less ambitious than all five targets, whereas the conditional pledges of most developing countries exceeded the average benchmark for a 2 °C limit. To address these challenges, Zenghelis (2017) advocated for a broader spectrum of approaches that leverage national interests and local co-benefits, pointing out that more than half of the mitigation actions required to achieve ambitious targets simultaneously generate local benefits.

These equity debates directly apply to shipping emission allocations. Heitmann and Khalilian (2011) included burden-sharing fairness among their core criteria. Selin et al. (2021) observed that flag-based allocation heavily burdens smaller nations while ship owners' income flows to industrialized countries, noting that states typically negotiate to minimize their own obligations. Deng and Mi (2023) highlighted the difficulty of attributing abatement responsibilities across complex, cross-regional maritime supply chains. Monios (2023) detailed IMO conflicts where developing nations like China, Brazil, and India cited common but differentiated responsibilities against

market-based policies, opposing the IMO's principle of equal treatment. Furthermore, no IMO mechanism accounts for historical emissions or the Global North's unequal gain from cheap transport.

Recent studies have sought to make equity measurable. In their research, Zhao et al. (2025) applied the Gini coefficient of national burdens to measure equity improvements, Wang et al. (2026) highlighted systemic inequalities between income groups, and Lu et al. (2023) designed allocation models that reconcile efficiency with fairness. Furthermore, Triantoro et al. (2026) examined these issues from a developing nation perspective, revealing long-standing tensions among climate-governance standards, difficulties in assigning cross-jurisdictional responsibility, and deficiencies in finance, technology, and institutions of these countries.

These concerns converge on the IMO Net-Zero Framework, approved in April 2025 by the Marine Environment Protection Committee. This framework combines the mandatory fuel intensity limits with GHG pricing for planned adoption in October 2025 and implementation in 2028 (United Nations Conference on Trade and Development [UNCTAD], 2025). However, in October 2025, the IMO postponed adoption to October 2026 following US agency opposition (Kang, 2026). For Kang (2026), the one-year delay in adopting the framework offers an opportunity to address key challenges. These include establishing reliable default values for well-to-tank emissions in life cycle assessment guidelines, overcoming the shortfall in zero- and near-zero-emission fuel supply, finalizing the transition fund structure and carbon pricing revenue rules for developing and least developed nations, and creating incentives for early adopters. Resolving these issues, the author notes, could position shipping as a model for achieving Paris Agreement targets. Each of them ultimately turns on how responsibility, and the financial obligation that follows from it, is distributed among states, which is why the question Selin et al. (2021) set out to answer, how alternative allocation choices redistribute national responsibility for shipping emissions, remains as pertinent today as when the allocation debate began.

One of the main restrictions for accurate allocation estimates is the access to activity and registry data, which are mostly held by commercial providers, and the cost of access has kept the allocation question in the hands of a small number of research groups. The consequence is that the findings of Heitmann and Khalilian (2011) and Selin et al. (2021), on which much of the policy debate rests, have not been independently reproduced, and cannot be updated as the fleet, the trade network, and the IMO's own targets continue to change. As Miola and Ciuffo (2011) highlight, "for a policy strategy to regulate air emissions to be effective, it must be based on robust estimates in terms of quantification and location."

This paper addresses that constraint directly. Our objective is to establish whether the method of Selin et al. (2021) can be reconstructed end to end from openly accessible data, and to document the reconstruction in enough detail that another researcher can carry it out. Thus, we describe the full pipeline, from vessel activity records to allocated national emissions, identify an open substitute for each database the original study relied on, and state explicitly where a substitution introduces a loss of coverage or precision that a user would need to account for. We then demonstrate the workflow on two vessels, COSCO ITALY (IMO 9516454) and RCC AMERICA (IMO 9277802), tracing each through every step to verify that the procedure is executable and that its intermediate quantities behave as expected.

We produce a validated and reproducible methodology, not a new global inventory. For this reason, the present analysis cannot support any claim about which allocation option distributes responsibility most effectively or most equitably. Scaling the procedure to the full fleet, and revisiting the allocation question with it, is the work this paper is intended to make possible. The remainder of the paper is organized as follows. Section 2 traces the evolution of emissions estimation methods, allocation options, and the data sources underlying both. Section 3 describes the

open-data reconstruction procedure step by step. Section 4 reports what the two-vessel demonstration reveals about the feasibility of the procedure and the sensitivity of its outputs to those substitutions.

2. Literature Review

2.1. Methods to estimate international shipping emissions

Estimating the emissions to be allocated is itself a methodological challenge. As Deng and Mi (2023) summarized, "to calculate ship CO_2 emissions, there are two main approaches that use different methods or models: bottom-up (activity-based) approach and top-down (fuel-based) approach" (p. 4). The authors found a 20% variation in CO_2 emissions reported across inventories due to differences in methodology and scope, with top-down approaches such as the IEA advancing timeliness while bottom-up approaches such as CAMS-GLOB-SHIP and EDGAR facilitate geospatial detail. Miola and Ciuffo (2011) provide a useful classification of studies by approach, present at the time of their work.

Early estimates relied on aggregated statistics. For instance, Heitmann and Khalilian (2011) calculated fleet emissions from annual average fuel consumption per ship type multiplied by the corresponding emission factors, a necessity given that "ships are neither obliged to report their route of navigation nor their consumption of fuel during their voyages" (p. 683). The Fourth IMO GHG Study applied both approaches, producing bottom-up estimates from ship activity data and top-down estimates from fuel sales statistics, with an uncertainty analysis of each. It also introduced a voyage-based allocation of international shipping, under which international CO_2 emissions rose from 701 to 740 million tonnes between 2012 and 2018 while remaining roughly 2% of the global total (IMO, 2020).

Between these two poles sit the activity-based models that differ mainly in how a position report is converted into engine power. Johansson et al. (2017) applied the third version of the Ship Traffic Emission Assessment Model to 7.8 billion AIS messages for 2015, predicting power from first principles by computing the frictional and residual resistance of every hull with a method fitted to tank tests of 433 hull forms (Hollenbach, 1998, as cited in Johansson et al., 2017) and converting that resistance into propelling and then brake power through a quasi-propulsive coefficient, with technical entries missing from the register supplied by a most-similar-vessel search over length overall and design speed rather than by ship-type averages. Their inventory covered 376,219 vessels and placed global shipping CO_2 at 832 million tonnes, a figure the authors treat as conservative because waves, currents and ice cover were excluded, effects they estimate at 5% to 15% of global fuel consumption. Schwarzkopf et al. (2021) took the opposite position on complexity with the modular ship emission modeling system, which derives the main-engine load from the cube of the ratio of instantaneous to service speed and obtains the characteristics that this requires from type-specific regressions fitted to more than 140,000 vessels, estimating gross tonnage from length or draught, installed main-engine power from the square root of gross tonnage, auxiliary power as a fixed fraction of installed power, and service speed as an asymptotic function of it. Validated against the Ship Traffic Emission Assessment Model over the North and Baltic Sea, it returned 10% to 18% less propulsion power yet closely similar CO_2, because a higher specific fuel consumption offset the lower power.

The study that serves as a basis in this work belongs mostly to the bottom-up methodology. Selin et al. (2021) calculated "CO_2 emissions as a function of engine power demand, activity time, and emissions factor" (p. 3), using an integrated dataset of terrestrial and satellite Automatic Identification System (AIS) data together with a global ship parameter database. Their inventory covered 433,940 ships in the 2015 AIS database, of which 69,399 had a valid and unique IMO number, and distinguished international from domestic voyages based on the geographical

range of ship activity, a distinction not explicitly made in previous studies. It is worth noting that the bunker fuel country option uses fuel-sales-based IEA data.

2.2. Methods to allocate international shipping emissions

Using 2007 data, Heitmann and Khalilian (2011) quantified the burden sharing implied by allocations based on bunker fuel sales, flag registration, operating company, and owning company, evaluating each against environmental effectiveness, ease of legal implementation, and fairness of burden sharing. The authors found that approximately 50% of the emissions would be allocated to the top ten countries in all four scenarios. They also concluded that flag-based allocation "would be environmentally ineffective both under the Kyoto and Copenhagen Accord scenarios because most shipping emissions would fall under flag states that have no commitments" (p. 687). No single option simultaneously fulfilled the three criteria, although they determined that allocation by operating company was the most defensible. The authors acknowledged that their conclusions rested on a single year of data and that "the use of ship movement data would be preferable to assess the actual miles that a country's fleet travelled in one year. However, at present, such data is not affordable for researchers" (p. 689).

After noting that "previous analyzes of national allocation schemes were limited by data availability on the national location of industry actors and inventories of CO_2 emissions from ships" (Selin et al., 2021, p. 2), the authors responded directly to that limitation by combining ship movement and bunker fuel sales data. With these data, they analyzed five national allocation options: flag country, owner country, manager country, operator country, and bunker fuel country, detailed in Table 1.

Their central result is distributional, meaning that "all allocation options would distribute a majority of CO_2 emissions to a small number of countries" (p. 4), with the top ten countries receiving between 69% and 75% of emissions and the top 20 between 88% and 90% under every option. The identity of those countries, however, varied sharply. Under the flag country option, the carbon budget of the Marshall Islands would increase by 51,203% and Liberia's by 8,143%, while Greece and Singapore could see increases of up to 108% and 232%, respectively. The analysis also exposed the interdependence of the options. For example, 62% of CO_2 emissions from Japanese-owned ships are allocated to Panama under the flag option, whereas 72% of emissions assigned to China under that option come from Chinese-owned ships. Weighing these outcomes against institutional criteria, the authors found that the mechanism best meeting effectiveness and equity criteria would assign emissions to the countries of ship owners and would operate under the UNFCCC (Selin et al., 2021).

Table 1. Definitions based on Selin et al. (2021) and SBSTA (1996). Equity implications summarized from Selin et al. and related discussion in this review.

Allocation option	Definition	Key actor	Equity implication
Flag country	Country where the ship is officially registered and flies its flag	Flag state is legally responsible for ensuring ships meet international standards	Heavy burden on small island states using flags of convenience (e.g. Marshall Islands: +51,203% carbon budget)
Owner country	Country where the entity that financially owns the ship is legally registered	Ship owner makes capital investment decisions on vessels and engine technology	Burden concentrates in OECD countries with capacity to finance decarbonization; recommended by Selin et al. (2021)
Operator country	Country where the entity that makes commercial decisions	Ship operator decides routing and how the asset is employed commercially	Similar OECD concentration to owner option; recommended by Heitmann & Khalilian (2011)

	on ship employment is registered		
Manager country	Country where the entity responsible for the daily running of the vessel is registered	Ship manager controls day-to-day operations including speed and fuel purchasing	OECD-concentrated; management often co-located with ownership, reducing differentiation from owner option
Bunker fuel country	Country where the marine bunker fuel seller is located, typically near major ports	Bunker fuel seller has no direct control over ship technology, routing, or speed	Non-OECD countries bear majority; large burden on port hubs (e.g. Singapore: +232%, Malta: +281%)

More recent work has expanded the allocation problem beyond these actor-based categories by linking maritime emissions to trade flows, voyage lifecycles, freight consumption, and shipping-network structure. These approaches show how the field has moved toward more granular and policy-specific ways of assigning responsibility. The following studies are therefore reviewed as alternative allocation frameworks that complement the Selin-style national allocation options.

Fry et al. (2024) imputed global freight routes with the bottom-up method and combined emissions with a multiregional input-output model, estimating that freight transport generated 2.8 Gt CO_2-equivalent in 2012, which corresponds to 41% of transport emissions and 9% of national footprints. They advocated for consumption-based allocation over bilateral trade shares while noting route imputation needs probabilistic uncertainty estimation. Wang et al. (2026) analyzed 1995–2021 exports, identifying a shift from high-income dominance to a balanced distribution, a 33.6% drop in global GDP-normalized emission intensity, and persistently high per capita emissions in high-income nations, which produce over half of global emissions with 15% of the population. They recommended a differentiated, multi-metric framework for equitable allocation.

Other studies examine voyages and networks. Zhao et al. (2025) linked crude oil tanker emissions to trade flows using a First-In-First-Out cargo-tracking framework, allocating 108 million tonnes of CO_2 across 2,491 tankers in 2020. Their approach lowered the trade-normalized Gini coefficient from 0.232 to 0.151, with country rankings remaining over 95% stable across varying exporter-importer factors, though 17.7% of emissions lacked complete loading-discharging cycles for allocation. Similarly, Lu et al. (2023) modeled offshore emissions across ~1,700 container shipping lines (98% of global lines, 2015–2020), finding central network dominance by developed and major economies and advocating greater responsibility for central nations. Tian et al. (2024) applied a bi-level programming model to liner shipping, proposing shortest-sailing-distance allocations per container to boost social benefits, enhance customer fairness, and lower dependence on self-reported data. Overall, recent research confirms that allocation choices fundamentally shift national responsibilities, echoing foundational literature.

2.3. Data sources

The data underlying this research topic has evolved together with its methods, and the sources involved fall into four broad groups: ship activity data, ownership and registry data, fuel and emissions statistics, and trade and network data. The paragraphs that follow describe each group of sources in turn, the studies that rely on it, and the degree to which it is openly accessible. Figure 1 below is a good illustration of the way Selin et al. (2021) applied the data from their different sources and the purpose each one served.

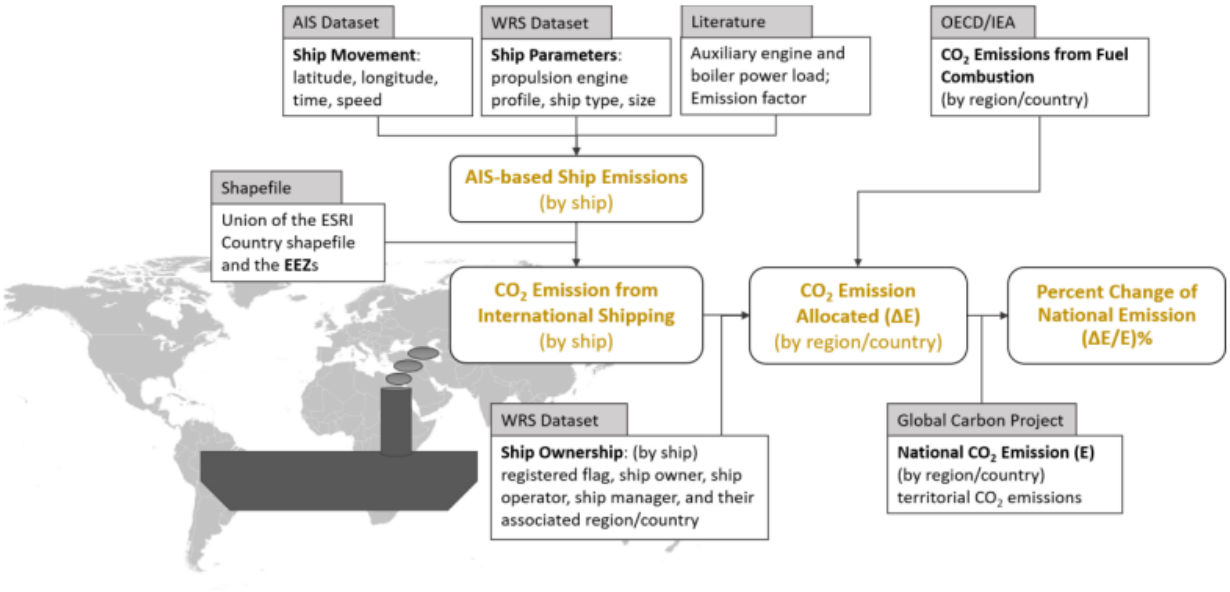


Figure 1. Methods diagram from Selin et al. (2021) supplementary material.

Ship activity data is the input that has since transformed the field. Selin et al. (2021) built their inventory from an integrated dataset of terrestrial and satellite AIS records for 2015. AIS data has allowed researchers to quantify emissions at high spatial-temporal resolutions with their real-time logs, including in open sea (Zhang et al., 2019). The Fourth IMO GHG Study aimed to update the global inventory of GHG emissions from shipping between 2012 and 2018, including, for the first time, estimates of carbon intensity and a voyage-based allocation of international emissions, and derived its bottom-up estimates from the same class of AIS-based ship activity data (IMO, 2020). More recently, the objective of Clarke et al. (2023) was to provide a new source of data for monitoring CO_2 emissions from global shipping that helps national statistics producers compile their air emission accounts, and their method is based on the granular and timely ship-level data provided by the United Nations Global Platform, producing results broken down by country and ship type that are available monthly and in near real time through the OECD's experimental database.

Access to this class of data is uneven, however. The integrated AIS dataset used by Selin et al. (2021) is maintained by commercial providers, the UN Global Platform restricts access primarily to national statistics offices, and the Fourth IMO GHG Study publishes aggregates suitable for validation, but not ship-level records.

Ship activity data alone cannot support a national allocation because it carries no information on who owns, manages, or operates a vessel, or on where it is registered. For this reason, Selin et al. (2021) complemented their AIS records with "data on the country where a ship is flagged as well as the national location of ship owners, operators, and managers from the World Register of Shipping (WRS) operated by IHS Maritime & Trade" (p. 3), a commercial registry now succeeded by Sea-web.

The completeness of such registries is itself a source of error. Zhang et al. (2019) set out to determine whether vessels that cannot be identified in ship parameter databases matter for AIS-based emission inventories, matching AIS records for the Pearl River Delta in 2015 against the WRS, which contains records of all vessels with registered

IMO numbers. The parameters missing for unmatched vessels were estimated through categorical regression using the ship type and length available in the AIS data itself. The authors found that these unidentified vessels accounted for almost 49% of overall ship CO_2 emissions in the region and, for this reason, strongly recommended that information on vessels without internationally registered identities be collected and publicly shared. Open substitutes for the commercial registries exist but are coarser, as detailed in Table 2. Equasis (2026) provides free ship-level flag, owner, and manager information, while UNCTAD publishes only country-level aggregates of fleet ownership and registration.

A third group of sources covers fuel consumption and national emissions statistics, which serve both as an alternative estimation route and as a benchmark against which bottom-up inventories are validated. Heitmann and Khalilian (2011) built their entire fleet inventory on this class of data. Selin et al. (2021) used bunker fuel sales data both as an allocation criterion in its own right and as a point of comparison, while the data on national CO_2 emissions and CO_2 emissions from fuel combustion also used in their research are available from the Global Carbon Project (2025) and the IEA (2018). Where the IEA series is not openly accessible, the United Nations Statistics Division energy statistics database reports bunker fuel deliveries by country, although it is not itself an emissions dataset.

A fourth group of sources links emissions to the trade they carry and draws on network and input-output data. The aim of Fry et al. (2024) was to calculate national carbon footprints that are complete with respect to freight transport emissions by allocating them to final consumption. The authors imputed international freight movements by transport mode from the bottom up and integrated the estimated emissions into a hybrid variant of the Eora multiregional input-output database, which links emissions to the trade in commodities and services between countries (Lenzen et al., 2013). Lu et al. (2023) obtained container shipping lines from the website of Alphaliner under a commercial license. This was combined with openly accessible sources, such as, container throughput data from the World Bank and geographical distances from the CEPII databases. Complementarily, UNCTAD's Review of Maritime Transport, a United Nations publication, provides the seaborne trade and fleet statistics on which much of the field relies (UNCTAD, 2025).

A recurring theme across all four groups is the tension between coverage and access. Statistics from the World Bank, CEPII, UNCTAD, the OECD, and the IMO studies are open to any researcher, whereas the richest activity data remain proprietary, including AIS feeds, vessel registries such as the WRS, and commercial line schedules. Additionally, gaps in these registries propagate directly into both estimation and allocation results (Heitmann & Khalilian, 2011; Zhang et al., 2019). Table 2 sets out, for each use, the source employed by Selin et al. (2021) and the public alternatives on which a replicable reconstruction could rely instead.

Table 2. Data sources used in Selin et al. (2021) and other similar sources. In bold, are the data sources that we used for the reconstruction methodology of this paper.

Use	Source	In Selin et al. (2021)	Access category	Role in this project	Notes / replication limitation
Ship activity data for emission estimates	S&P Global Market Intelligence. AISLive ship tracking / Sea-web AIS (source)	X	Commercial/proprietary	Original Selin input	Timestamped vessel positions, speed, and draught used to estimate activity and fuel burn. Exact replication is not possible without licensed access.
	Global Fishing Watch. AIS presence, vessel identity, and port visits (source)		**Open data/API**	**Main open substitute**	**Provides hourly AIS presence/positions and port-visit events. Coverage, vessel filtering, and identity fields may differ from Selin's AIS dataset.**
	United Nations Statistics Division. AIS data, UN Global Platform (platform; handbook)		Restricted institutional access	Not used as main input	Potential AIS source, but access is mainly for NSOs, established partners, and UN community members. Not fully reproducible for independent researchers.
	Spire Maritime AIS (source); Kpler / MarineTraffic AIS API (source)		Commercial/proprietary	Possible commercial substitutes	Similar class of AIS movement data.
Emissions model constants and validation totals	**International Maritime Organization. Fourth IMO GHG Study 2020 (source)**		**Open report / validation only**	**Model constants and validation reference**	**Used for operating modes, auxiliary/boiler loads, fuel consumption assumptions, CO_2 emission factors, and aggregate validation. Not ship-level activity data.**
Vessel specifications and allocation keys	S&P Global Market Intelligence. Sea-web maritime reference (source)	X	Commercial/proprietary	Original Selin input	Successor to WRS. Includes ship characteristics, flag, owner, operator, manager, engine power, design speed, tonnage. Exact field matching requires licensed access.
	MarineTraffic: Global Ship Tracking Intelligence (source)		**Free registration. Options to upgrade for full access at a cost**	**Ship LOA for power estimation**	**Provides ship-level flag, type, moulded beam, draught and LOA.**
	Equasis. Ship database (source)		**Free registration**	**Main open substitute**	**Provides ship-level flag, DWT/GT, registered owner, ISM manager, and commercial manager. Commercial manager is used as an operator proxy. No bulk download, definitions may not perfectly match WRS.**
Fleet ownership / registry aggregates	UNCTADstat. Merchant fleet and beneficial		Open aggregate data	Validation / context only	Country-level ship counts and DWT by flag or ownership location. No individual vessels,

	ownership (fleet by flag; beneficial ownership)				operator, or manager fields. Not suitable for ship-level attribution.
Missing vessel technical parameters	**Cepowski & Chorab (2021); US EPA vessel-power regressions**		**Open literature / public methods**	**Supporting estimates**	**Used where free vessel specifications are unavailable. Introduces model-based uncertainty and should be reported separately from observed Equasis specifications.**
Installed power and reference speed estimates	**IMO Resolution MEPC.333(76) (source)**		**Open regulatory document**	**Supporting estimation method**	**Provides EEXI-based reference-speed and installed-power estimation logic. Used to fill technical gaps, not as a direct activity dataset.**
Geographical boundaries of countries, EEZs, high seas, and ECAs	**Marine Regions / VLIZ. Maritime boundaries geodatabase (source)**	**X**	**Open geospatial data**	**Main spatial classification input**	**No substitution needed. Use World EEZ v12, High Seas v2, MARPOL ECA polygons, and Marine/Land Zones v4 for spatial classification.**
International maritime CO_2 emissions per country	International Energy Agency. World Energy Balances / GHG Emissions from Energy (source)	X	Commercial/proprietary or restricted data product	Original Selin bunker-fuel source	Selin used IEA international maritime CO_2 emissions by country for the bunker fuel country option.
International maritime bunker-fuel allocation	United Nations Statistics Division. UNdata Energy Statistics (fuel oil; gas oil/diesel oil)		Open aggregate data	Fleet-scale bunker-fuel allocation option	Retained for fleet-scale analysis, not used in the two-vessel pilot. Not an emissions dataset. CO_2 must be derived from bunker fuel deliveries using emission factors.
Changes to national carbon budgets	**Global Carbon Budget 2025 (source)**	**X**	**Open aggregate data**	**Country-level territorial CO_2 baseline**	**Selin used the 2018 release. This project uses the 2025 release to contextualize allocated shipping emissions against national territorial CO_2.**
Territory alignment for country baselines	**Selin et al. (2021), supplementary Table 1 (article)**	**X**	**Open supplementary material**	**Replication alignment rule**	**Used to harmonize territories and countries following Selin's treatment.**

3. Methodology

The seven-digit IMO number is the stable key throughout, because names, MMSI numbers, flags, and provider identifiers all change over the life of a hull while the IMO number does not (IMO, 2016). The overall logic follows Selin et al. (2021), who calculated CO_2 as a function of engine power demand, activity time, and emission factor. Table 3 sets out the inventory the reconstruction runs on. No single source describes a complete ship, so activity, identity, hull geometry, engine parameters, spatial boundaries, and national baselines are assembled from five separate families and joined on the IMO number.

Table 3. Data sources, the parameters taken from each, and the calculation in which they are used.

Data source	Parameters retrieved	Calculation in which they are used
GFW 4Wings presence report (public-global-presence)	Hourly latitude and longitude, hours credited per cell, and inline identity (IMO number, name, flag, ship type)	Hourly track; derived speed over ground, equations (B1) to (B3); smoothing, equation (1); coverage and gap classification, Section 3.1
GFW events, port visits (public-global-port-visits-events)	Visit start and end times, anchorage identifier and coordinates, port country code, dock flag, confidence level, duration	Voyage legs and their international, domestic and EU-to-EU labels, Section 3.2; the at-berth mode of Table 5; the out-of-service test of Section 3.1
GFW vessels identity endpoint	GFW vessel identifiers associated with an IMO; registry and self-reported identities from which candidate names are reconstructed	Retrieves port visits for every associated vesselId; helps configure exact historical ship-name filters. Presence observations themselves are validated against the expected IMO
Equasis	Flag, gross and deadweight tonnage, ship type, year of build, registered owner, ISM manager and commercial manager, with company addresses	Size band and engine assumptions, Section 3.3.
Marine Regions World EEZ v12 and World High Seas v2	EEZ polygons and sovereign-state codes; high-seas polygon retained as a not-in-any-EEZ reference layer	EEZ intersection supports the whole-ship domestic diagnostic. In the current implementation, hours unmatched to EEZ are classified as high seas; voyage-leg labels, not EEZs, determine allocation
Marine Regions MARPOL Annex VI Regulation 14 layer	Six emission control area polygons	Distillate and residual fuel assignment, Section 3.5
Marine Regions Marine and Land Zones v4	Combined land and zone polygons, from which the coastline is derived	Distance to coast in the mode matrix, Table 5

3.1. Ingest and preprocess AIS data

This step produces, for every ship and every hour of the study period, a position and a speed. Selin et al. (2021) built this layer from roughly 896 million terrestrial and satellite AIS messages for 2015 supplied by a commercial provider. That dataset is not openly available, so we substituted the Global Fishing Watch presence product (Global Fishing Watch, 2025), which is free for non-commercial research and covers all vessel types globally.

For every vessel and calendar year, the pipeline submits a single query to the GFW 4Wings presence report API using the public global presence dataset. The request demands high spatial resolution with hourly temporal granularity, grouped by vessel identifier, and targets a world-extent polygon, a near-global bounding box covering 180°W–180°E and 85°S–85°N. Because *shipname* matching is exact, case-sensitive, and not inherently unique, each API call incorporates all recorded historical names within a single *shipname in (...)* filter. Since GFW presence data cannot be directly filtered by IMO, the loader validates that returned non-null IMO values correspond strictly to the configured, checksum-valid IMO before retaining those records. The GFW Vessels endpoint is the API route for vessel identity/registry information.

Each annual pull needs to fulfill the following criteria:

- it must be non-empty,
- it must contain exactly the configured IMO number and that number must pass the check-digit test,
- it must cover at least 10% of the elapsed hours of the year, and
- it must not exceed them by more than 1%.

A year below 95% observed coverage is retained but flagged as low confidence. These criteria exist because the GFW endpoint returns an empty rather than an error response when a name is misspelled, so a silent failure is otherwise indistinguishable from sparse activity. Where several high-resolution cells belong to one vessel-hour, the positions are collapsed to an hours-weighted centroid.

Selin et al. (2021) retrieves the speed from the AIS dataset. However, speed over ground is not recoverable from the GFW dataset. Thus, we construct a centroid-based SOG proxy from consecutive positions by the haversine construction given in Appendix B, equations (B1) to (B3). Doing so introduces a bias. Each vessel-hour is assigned to a single grid cell, so a ship actually cruising near 15 knots, can appear to jump from 3.36 to 21.63 knots, for example. Because propulsion power varies with the cube of speed, so unsmoothed hourly speeds overstate main-engine power by a factor of 1.94 on the container hull and 2.06 on the vehicle carrier, which centred averaging reduces to 1.26 and 1.34 at a seven-hour window (Table 9). The pipeline therefore applies the centred moving average of equation (1) and treats the window length as a parameter to be tested.

$$SOG_{w,i} = \frac{1}{w}\sum_{k=-(w-1)/2}^{(w-1)/2} SOG_{raw,i+k} \quad (1)$$

$SOG_{w,i}$ is the smoothed speed at hour i for a window of w hours, w takes the odd values 1, 3, 5, and 7, and k indexes the hours inside the window. A window is applied only within a contiguous underway segment and never crosses a port visit or an out-of-service interval. Port-visit speeds are left unsmoothed, and inactive hours are null.

The pipeline constructs a complete hourly record, filling missing positions by nearest neighbour and derived SOG values by linear interpolation between bracketing observations. At the record boundaries, it uses one-sided filling where no bracketing observation is available. Every filled hour is flagged. This adopts Selin et al. (2021)'s interpolation methods, but not their daily eligibility rule. The authors interpolated only when a ship had at least one AIS signal that day, whereas this pipeline fills all non-inactive missing GFW hours. Then, their reported 32% interpolation share is not directly comparable.

Extended missing periods have a different classification process. Drawing from GFW Events data, a port call is defined as a vessel's recorded visit to a port, spanning from its initial detected arrival time to its subsequent departure. When a data gap persists for seven or more consecutive days without any port call within that window, it is designated as out of service. Conversely, if a port call begins during a prolonged gap, the system flags an error

instead of treating the duration as inactive. Periods categorized as inactive are assigned zero emissions. Figure 2 depicts a single month of tracking data for the container ship, highlighting the distinction between observed hours and interpolated timeframes.



Figure 2. Monthly infilled speed over ground for January 2019.

3.2. Consolidate ship movement

Allocation needs to know not only where a ship was but whether the hour belonged to an international voyage. Port visits are downloaded from the GFW events API at confidence levels three and four, for every ship identifier associated with the IMO number, and are combined and deduplicated by event identifier. Confidence levels are provided by GFW and indicate confidence in the inferred port visit. This is one difference with the AIS database of the original paper, GFW port events are inferred, so they may split or merge a physical port stay differently from the raw-AIS detection algorithm of the Fourth IMO Greenhouse Gas Study.

Each call records its start and end times, its start and end anchorages, the port-country code, the anchorage coordinates, a dock flag, the confidence level, and the first EEZ MRGID reported for the event, which is a Marine Regions' unique numeric identifier for an Exclusive Economic Zone polygon.

A voyage leg runs from the end of one port call to the start of the next. Its origin country is taken from the previous call's end anchorage and its destination country from the next call's start anchorage. A leg is international when its two port countries differ and domestic when they agree, and it is left unlabelled when either country is missing. A

leg is additionally marked as internal to the European Union when both port countries are members, with the United Kingdom included through 2020, a label used by the fuel rule of Section 3.5. This voyage-based definition implements Option 2 of the Fourth IMO Greenhouse Gas Study, under which a voyage is international when it connects ports in two different countries (IMO, 2020, p. 60).

Selin et al. (2021) classified a whole ship as domestic when more than 95% of its annual AIS signals fell within a single EEZ and classified all other ships as international for the entire year. The pipeline retains an adapted version of this rule as the equation (B5) diagnostic. A vessel is domestic when more than 95% of its non-inactive hourly record, including high-seas hours in the denominator, lies within one sovereign EEZ. Because this diagnostic uses interpolated GFW hours and excludes out-of-service hours, it is not a direct reproduction.

For this diagnostic, non-inactive vessel-hour positions are intersected with World EEZ version 12 polygons from the Flanders Marine Institute (2023). The sovereign-state field is used, unmatched positions are treated as high seas, and joint-regime or overlapping-claim matches are flagged as disputed.

Figure 3 shows these zone layers together with the recorded positions of the two pilot vessels.

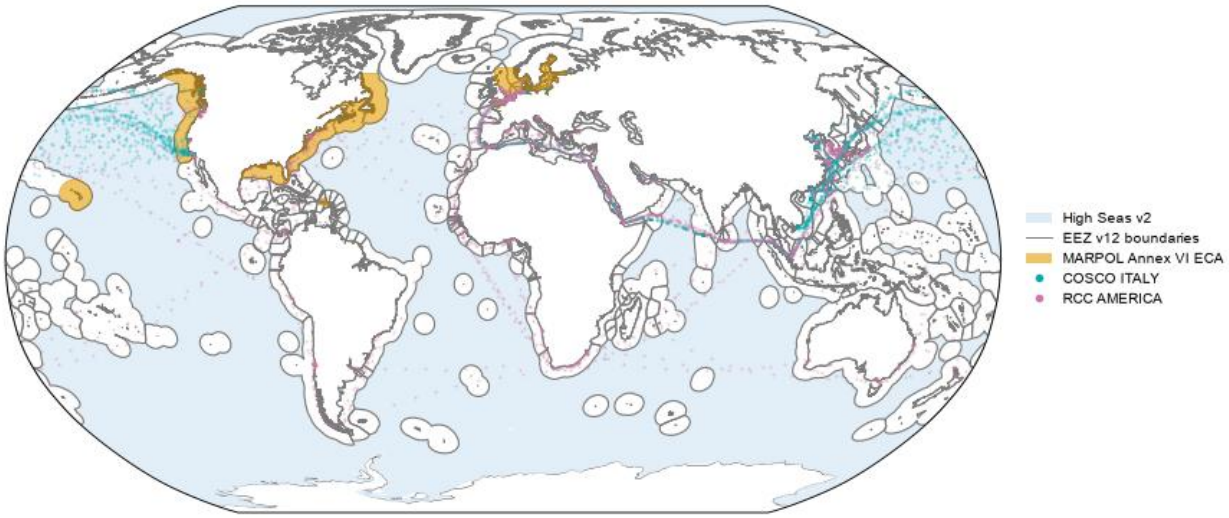


Figure 3. Marine Regions datasets and pilot-vessel activity.

3.3. Acquire ship specifications

As Zhang et al. (2019) explained it, inventory estimation requires not only ship activity data but also pertinent ship parameters such as engine power that are not included in the AIS activity database. Selin et al. (2021) obtained flag, owner, operator, and manager countries together with maximum main-engine power, design speed, ship class, capacity, and auxiliary characteristics from the World Register of Shipping operated by IHS Maritime and Trade, a subscription product. The public substitute is Equasis, supplemented where necessary by public registry records. For the pilot vessels, these sources provide, as available, flag, registered owner, ISM manager, commercial manager, GT, DWT, ship type, build year, and dimensions. They do not publish engine power, design speed, and an operator field.

That absence is the largest gap between the public and the commercial data. Selin et al. (2021) met a smaller version of the same problem and applied the categorical regression of Zhang et al. (2019) to the vessels their register did not cover. The same solution is not available here because no ship is covered. The fallback regressions of the Fourth IMO Greenhouse Gas Study were also rejected, because their published coefficients are circular for a ship whose installed power is unknown. Other studies and guidelines present approaches that we have implemented in this methodology. Our methodology defines four estimates (A–D), without selecting a primary estimate. Estimates A, B, and D are executable with the available inputs, subject to vessel-specific restrictions, while Estimate C is deliberately reserved for a sourced vessel-specific specification and raises an error until those inputs are supplied.

The Fourth IMO Greenhouse Gas Study indexes container ships by capacity in twenty-foot equivalent units, which Equasis does not report, so capacity is estimated from the moulded beam using the preliminary-design formula of Cepowski and Chorab (2021), given as equation (B6). Both vessels are assigned a slow-speed diesel main engine, the default that study prescribes for an oil-propelled ship that cannot otherwise be classified.

3.3.1.Estimate A

Estimate A applies the average reference curves published with the 2021 Guidelines on the attained Energy Efficiency Existing Ship Index (IMO, 2021). For each of twelve ship categories, the curves estimate average reference speed and average installed main-engine power from a capacity measure, as shown in equations (2) and (3).

$$V_{ref,avg} = A\,B^{C} \tag{2}$$

$$MCR_{avg} = D\,E^{F} \tag{3}$$

$V_{ref,avg}$ is a statistical mean of distribution of ship speed in given ship type and size, MCR_{avg} is a statistical mean of distribution of original installed power for main engines, A, C, D, and F ship-category coefficients tabulated in the resolution, and B and E the capacity measures of the category (IMO, 2021, 2.2.3.5 and appendix). For containerships, B = min (DWT, 80,000) for speed and E = min (DWT, 95,000) for power. These caps bind for the container-ship pilot and must not be omitted.

3.3.2.Estimate B

Estimate B applies the Admiralty relationship of Molland (2008), which holds for geometrically similar hulls at equal Froude number. A reference speed follows from a Froude number and the length between perpendiculars in equation (4), which inverts Molland (2008, equation 5.5), and the installed power from the displacement and that speed in equation (5), which rearranges his Admiralty coefficient to solve for power.

$$V_{ref} = \frac{F_n\sqrt{g\,L_{BP}}}{0.5144} \tag{4}$$

$$MCR = \frac{\Delta^{2/3}{V_{ref}}^{3}}{C_{adm}} \tag{5}$$

V_{ref} is the reference speed in knots, F_n the Froude number, g gravitational acceleration, L_{BP} the length between perpendiculars in metres, MCR the installed main-engine power in kilowatts, Δ the displacement in tonnes, and C_{adm} the Admiralty coefficient, whose median of 482 and range of 352 to 593 are back-calculated for this study from the 17 container ships of Charchalis (2014), Table 1, as derived in Table C4. “This proportionality is termed the

Propeller law, and despite it being termed a law that the required power is proportional to the cubic of the speed, it is only an assumption applicable at lower Froude numbers" (MAN Diesel, 2018, p. 27).

Molland defines the coefficient on shaft power, which equals brake power in a direct-drive installation, so equation (5) returns installed rated power because C_{adm} is calibrated against installed power, which is why the reference load fraction of Table 4 is 1.0 here. The unit fraction is retained because equation (5) pairs speed and installed power exactly as Charchalis (2014) tabulates them, so the estimate is internally consistent on its own terms. It does assume that the tabulated speed is reached at full installed power, whereas a service speed is normally attained below it, at the 85 to 90% normal continuous rating reported by Molland (2008) or the 75% for Estimate A, so Estimate B is anchored higher than the other routes and its main-engine demand is best read as an upper bound. Assigning it a fraction in that band, 0.83 to match Estimate D, would make the three routes like-for-like and lower its main-engine demand by roughly a sixth, and is the first refinement to make if the comparison is tightened.

The Froude number range is taken from Cepowski and Chorab (2021), and displacement from the two conventions of Charchalis (2014), a geometric calculation from block coefficient, hull dimensions and seawater density, and a deadweight-to-displacement ratio. The point scenario uses the midpoint of the Froude-derived speeds and the mean of the two displacements, with the endpoint combinations retained as a range diagnostic. The calibration sample is 17 feeders at Froude numbers of 0.22 to 0.28 and 22,215 to 42,157 tonnes, so the coefficient is applied below its speed window and far above its size range. The restriction lies in the coefficient since C_{adm} varies with hull form, size and Froude number and none is published for pure car and truck carriers, hence Estimate B is evaluated for the container ship only.

3.3.3. Estimate D

Estimate D applies the regressions of the United States Environmental Protection Agency (US EPA, 2000), which predict rated main-engine power directly from deadweight tonnage in equation (6).

$$MCR = \alpha\, DWT + \beta \tag{6}$$

MCR is the rated main-engine power in horsepower, converted to kilowatts, *DWT* the deadweight tonnage, and α and β the slope and intercept of the relevant ship group. Because the regression predicts power alone, it is paired with a speed drawn from another estimate and does not bracket speed independently.

3.3.4. Estimate C

Estimate C is a vessel-specific sourced specification instead of a regression. It requires an independently reported installed main-engine maximum continuous rating (MCR), reference speed, and the reference condition that relates the two, for example, the main-engine load at the reported speed.

No consistent public source was identified for these parameters for both pilot vessels. That is why Estimate C is retained as a reserved scenario but is not included in the active calculations. A researcher with access to a shipbuilder, class-society, or equivalent vessel-specific record can populate these fields, with their source and method, and rerun the pipeline without changing the model code.

Table 4 gives the values these three estimates use for the two pilot vessels. The complete coefficient sets, covering every ship type, are reproduced in Appendix C.

Table 4. Values used by each specification estimate for the two pilot vessels.

Estimate	Quantity	COSCO ITALY (container)	RCC AMERICA (vehicle)	Source
A	Speed coefficients, A and C	3.2395 and 0.18294	16.6773 and 0.01802	IMO (2021), MEPC.333(76)
A	Power coefficients, D and F	0.5042 and 1.03046	262.7693 and 0.39973	IMO (2021), MEPC.333(76)
A	Capacity used, speed and power	80,000 t and 95,000 t, both capped	21,182 t, uncapped	Equasis, caps from IMO (2021)
A	Reference load fraction, fref	0.75	0.75	IMO (2021), 75% of original MCR
B	Froude number range, Fn	0.19 to 0.21	not calibrated	Cepowski and Chorab (2021)
B	Length between perpendiculars, LBP	347.60 m	189.82 m	0.95 times the length overall
B	Block coefficient, CB	0.61	not calibrated	Charchalis (2014), assigned
B	Deadweight to displacement ratio	0.80	not calibrated	Charchalis (2014)
B	Admiralty coefficient, Cadm	482, range 352 to 593	not calibrated	Back-calculated in Table C4
B	Reference load fraction, fref	1.00	not applicable	This study, see the text above
D	Regression slope and intercept	0.800 and -749.4 hp	0.719 and +2,581 hp	US EPA (2000), EPA420-R-00-002
D	Deadweight	156,610 t	21,182 t	Equasis, observed
D	Speed pairing	Froude speed of Estimate B	Reference speed of Estimate A	This study
D	Reference load fraction, fref	0.83	0.75	US EPA (2000); IMO (2021)

Estimate B is not calibrated for vehicle carriers, so the hull-form constants it requires have no published values for vehicle carriers. The reference speeds and installed powers these values produce are reported in Section 4.

3.4. Estimate engine power demand

Schwarzkopf et al. (2021) observed that ship emission models differ in complexity, and in particular in how they calculate a ship's power demands, with sophisticated concepts considering resistances caused by external influences while more simplistic approaches rely on engine load factors and characteristics of the ship and its engines. This model belongs to the second family, as does that of Selin et al. (2021), who expressed main-engine power as a function of maximum engine power, real-time speed, and design speed. The choice is dictated by the data, because the GFW exposes neither draught nor navigational status and no public source supplies a hull-specific speed-power curve, weather along the track, or hull condition.

Auxiliary demand depends on what the ship is doing rather than on how fast it moves, so each vessel-hour is assigned an operating mode before its auxiliary load is read. The pipeline implements Table 16 of the Fourth IMO Greenhouse Gas Study as an ordered decision matrix producing five modes, combining smoothed speed, distance to the nearest port, distance to coast, and main-engine load, as summarized in Table 5.

- Distance to port: measured to the nearest start or end anchorage in the vessel's own recorded call history, which keeps the measure consistent with the events that generate the calls, so it records the distance to a port the vessel actually entered and not the nearest port anywhere.
- Distance to coast: derived from the Marine and Land Zones version 4 layer of the Flanders Marine Institute (2024) by differencing each combined land and zone polygon with the matching zone polygon, giving 253 land polygons from 328 source features, so that distance is measured to the territorial-sea baseline and not to a physical shoreline, an approximation that touches 0.2% of the container ship's distinct positions, all of which already lie within a nautical mile of a port.
- Neither distance is computed beyond a narrow neighbourhood of a port or of land, and a position outside that neighbourhood carries no distance at all, which the matrix reads as lying beyond the 5 nautical mile limit at which Table 16 stops distinguishing.

We implement a specific modification to the operational phase logic defined in Table 16. In the case of vessels other than liquid tankers, the standard procedure distinguishes between the *at_berth* and *anchored* states solely by proximity, designating hours with speeds not exceeding 1 knot as being "at berth" only when located within 1 nm of a port. However, because coordinates for GFW anchorages frequently reside at a distance from the actual physical terminal, vessels effectively moored may be registered beyond this 1 nm threshold. For the container ship pilot we used, applying the proximity-based rule identified a mere 1,143 hours as *at_berth*, while GFW's inferred port-visit intervals captured 17,427 hours. This discrepancy resulted in roughly 16,000 hours being erroneously associated with the 1,800 kW auxiliary demand typical of the *anchored* state, instead of the 1,300 kW required for *at_berth* operations. To address this, any vessel-hour recorded at speeds ≤ 1 knot during a GFW-detected port stay is categorized as *at_berth*, irrespective of its distance.

Table 5. Operational phase assignment decision matrix.

SOG (knots)	ME load	Port ≤1 nm	Port 1 to 5 nm*	Coast ≤1 nm	Coast 1 to 5 nm	Coast ≥5 nm
≤1	any	At berth	At berth*	Anchored	Anchored	Anchored
1 to 3 (incl. 3)	any	Anchored	Anchored*	Anchored	Anchored	Anchored
3 to 5 (incl. 5)	≤ 0.65	Manoeuvring	Manoeuvring*	Manoeuvring	Manoeuvring	Slow transit
3 to 5 (incl. 5)	> 0.65	Manoeuvring	Manoeuvring*	Manoeuvring	Manoeuvring	Normal cruising
>5	≤ 0.65	Manoeuvring	Slow transit*	Slow transit	Slow transit	Slow transit
>5	> 0.65	Manoeuvring	Normal cruising*	Normal cruising	Normal cruising	Normal cruising

*Source: Table 16 of the Fourth IMO Greenhouse Gas Study (IMO, 2020, printed p. 66). The first applicable row is selected on speed, and within that row the first applicable column wins. * Applicable to chemical tankers, liquefied gas tankers, oil tankers and other liquids tankers only. The at-berth rule is modified as described in the text.*

The assignment is implemented in `40_operating_mode.sql` as the ordered conditional expression reproduced below, in which the first branch that matches determines the mode.

```
CASE
    WHEN h.sog <= 1 THEN
        CASE
            WHEN $use_port_visit_intervals AND v.in_port_visit THEN 'at_berth'
            WHEN d.port_nm <= 1 THEN 'at_berth'
            WHEN $is_liquid_tanker AND d.port_nm <= 5 THEN 'at_berth'
            ELSE 'anchored'
        END

    WHEN h.sog <= 3 THEN 'anchored'

    WHEN h.sog <= 5 THEN
        CASE
            WHEN d.port_nm <= 1 THEN 'manoeuvring'
            WHEN $is_liquid_tanker AND d.port_nm <= 5 THEN 'manoeuvring'
            WHEN d.coast_nm IS NOT NULL AND d.coast_nm < 5 THEN 'manoeuvring'
            WHEN h.me_load <= 0.65 THEN 'slow_transit'
            ELSE 'normal_cruising'
        END

    ELSE
        CASE
            WHEN d.port_nm <= 1 THEN 'manoeuvring'
            WHEN h.me_load <= 0.65 THEN 'slow_transit'
            ELSE 'normal_cruising'
        END
END AS operating_mode
```

Here `h.sog` is the smoothed speed of the scenario, `h.me_load` the calculated main-engine load fraction, `d.port_nm` and `d.coast_nm` the distances to port and to coast in nautical miles, and `v.in_port_visit` the flag marking the hours that fall inside a GFW port-visit interval, while `$use_port_visit_intervals` and `$is_liquid_tanker` are the configuration switches governing the two rules described above.

3.4.1.Main engine power

Main-engine demand follows from the ratio of the smoothed speed to the reference speed of the scenario, raised to the speed exponent, scaled by a reference load fraction and multiplied by the installed power, as shown in equations (7) and (8).

$$Load_i = min\left(1,\ f_{ref}\left(\frac{SOG_{w,i}}{V_{ref}}\right)^n\right) \tag{7}$$

$$W_{ME,i} = MCR \cdot Load_i \tag{8}$$

Loadi is the dimensionless load factor at hour i, f_{ref} the fraction of installed power at which the reference speed is attained, $SOG_{w,i}$ the smoothed speed from equation (1), V_{ref} and MCR the reference speed and installed power of the scenario, n the speed exponent of 3, and $W_{ME,i}$ the main-engine demand in kilowatts.

The cubic dependence is the propeller law, which Molland (2008) describes as an approximation valid over limited power ranges and whose formal speed sensitivity of three he derives in linearized form. Selin et al. (2021) specify the same three arguments of installed power, instantaneous speed and design speed, but state no formula and refer the description of their emission model to Zhang et al. (2019). The reference load fraction is necessary because the reference speed is not attained at full installed power. Resolution MEPC.333(76) defines the attained EEXI at 75% of the maximum continuous rating, the EPA ShipPowerModel places service speed at 83% of installed power, and

Molland (2008) reports normal continuous ratings of 85% to 90% of the maximum for propeller design together with owner requirements of 80% or less at service speed. Therefore, the fraction is assigned per estimate from the source that defines it: 0.75 for Estimate A, 0.83 for Estimate D, and 1.00 for Estimate B, whose Admiralty coefficient is calibrated against installed power rather than against a service condition. Load is capped at unity, and demand is set to zero below 7% load and in the at-berth and anchored modes. The 7% floor is the IMO low-load threshold, which is defined against an unscaled load factor, so applying it to the scaled factor raises the speed at which the main engine is zeroed.

The estimate reflects a vessel operating under reference loading conditions in calm water with an unfouled hull, and thereby underrepresents the energy required by a fully loaded ship navigating in adverse weather. The Fourth IMO Greenhouse Gas Study corrects for instantaneous draught relative to reference draught, for weather and for hull fouling, and the Admiralty form of the load factor carries the first of these as the ratio of instantaneous to reference draught raised to two thirds, but none of these adjustments can be implemented in the present framework because neither the loading state nor the sea state is observable in the activity data. The omission is not negligible, as Bialystocki and Konovessis (2016, p. 157) demonstrate that "stronger wind and head weather increases the fuel consumption, and the difference between several weather conditions could be quantified".

3.4.2.Auxiliary engine and boiler power

Auxiliary engines supply the electrical load, which depends on activity, so demand is read from Table 17 of the Fourth IMO Greenhouse Gas Study by a range join on ship type, size band, and operating mode, expressed jointly with the boiler lookup in equation (9).

$$\left(W_{AE,i},\ W_{BO,i}\right) = f(type,\ size\ band,\ mode_i) \tag{9}$$

$W_{AE,i}$ and $W_{BO,i}$ are the auxiliary engine and boiler demands at hour i in kilowatts, f denotes the tabulated lookup, *type* is the Equasis ship type, *size band* is the capacity band in twenty-foot equivalent units for container ships and in deadweight tonnage for the other types considered here, and $mode_i$ is the mode of hour i. The slow transit and normal cruising modes both read the sea column. Selin et al. (2021) determined auxiliary demand the same way, from ship class, capacity, and activity mode.

Below 150 kilowatts of installed main-engine power, auxiliary and boiler demand are both zero. Between 150 and 500 kilowatts, auxiliary demand is 5% of installed power while boiler demand is still read from the table. Above 500 kilowatts both are read as published. Table 6 reproduces the values for its size band and shows why the mode assignment matters. Auxiliary demand alongside the berth is about two-thirds of the demand at sea, while boiler demand is 630 kilowatts alongside zero at sea, so misclassifying port hours changes the annual total materially even though those hours carry no propulsion.

Table 6. Auxiliary engine and boiler power for the 12,000 to 14,499 TEU container band.

Operating mode	Auxiliary engine (kW)	Auxiliary boiler (kW)
At berth	1,300	630
Anchored	1,800	630
Manoeuvring	3,250	630
Sea	2,050	0

Source: Table 17 of the Fourth IMO Greenhouse Gas Study (IMO, 2020). The table covering 14 ship types and seventy size bands, is reproduced as Table C5 in Appendix C, and the small-engine overrides as Table C6.

3.5. Calculate CO_2 emissions

Ship emissions generate air pollutants such as SOx, NOx, PMs, CO and hydrocarbons (Zhang et al., 2019), but this study quantifies CO_2 alone, following Selin et al. (2021). The species inventory of Schwarzkopf et al. (2021) supports the restriction on mass grounds, since CO_2 accounts for roughly 98% of the total mass emitted by the global fleet and exceeds NOx, the largest non-CO_2 species, by a factor of about fifty. Restricting the inventory to CO_2 is a decision about scope, since SOx, NOx and particulate matter carry the health and acidification impacts that motivate MARPOL Annex VI and the emission control areas, and they enter this analysis only through the fuel sulphur limits that constrain fuel choice.

Fuel is assigned hourly per ship because both the specific fuel consumption of an engine and the CO_2 emission factor depend on the fuel burned. An hour is assigned marine diesel or gas oil (MDO) when the main engine is a high-speed diesel, when its position falls inside an emission control area designated under regulation 14 of Annex VI to MARPOL, or when it lies on a voyage leg between two European Union ports, and heavy fuel oil otherwise. This rule reproduces that of Selin et al. (2021), with one refinement. They worked from ship activity ranges, whereas the port-to-port legs of Section 3.2 allow the European Union condition to be tested on actual voyages rather than on a geographical proxy. An hour inside a port visit belongs to no leg and so is tested against the emission control area condition alone. The emission control area polygons are the regulation 14 layer published by the Flanders Marine Institute (2020), containing the Baltic Sea, the North Sea, the United States Caribbean, and the three North American areas. The Mediterranean area entered into force in May 2025 and so falls outside the study period, but a later study year must add it. The 2020 sulphur cap creates no branch in the CO_2 calculation, because heavy fuel oil and its low-sulphur variant carry the same carbon factor in Table 21 of the Fourth IMO Greenhouse Gas Study, although it would matter for sulphur or particulate emissions.

The rule is implemented in `30_fuel_assignment.sql` as the query reproduced below, in which the three distillate conditions are tested in order and the residual fuel is the fallback.

```
SELECT
    h.imo,
    h.ts,
    e.in_eca,
    e.eca_area,
    coalesce(l.is_eu_eu, FALSE) AS is_eu_eu_leg,
    $main_engine_is_high_speed AS main_engine_high_speed,
    CASE
        WHEN $main_engine_is_high_speed THEN $distillate_fuel
        WHEN e.in_eca THEN $distillate_fuel
        WHEN coalesce(l.is_eu_eu, FALSE) THEN $distillate_fuel
        ELSE $residual_fuel
    END AS fuel_type
FROM vessel_hour AS h
LEFT JOIN eca_hour AS e
       ON e.imo = h.imo
      AND e.ts = h.ts
LEFT JOIN voyage_leg AS l
       ON l.imo = h.imo
      AND h.ts >= l.depart_ts
      AND h.ts < l.arrive_ts;
```

Here `h.imo` and `h.ts` identify the vessel-hour, `e.in_eca` is the flag set by the point-in-polygon join against the emission control area layer, `l.is_eu_eu` marks a voyage leg whose departure and arrival ports both lie in the European Union, and `$main_engine_is_high_speed`, `$distillate_fuel` and `$residual_fuel` are the configuration values carrying the engine class of the vessel and the two fuel labels, while the left join to `voyage_leg` returns no row for an hour inside a port visit, which `coalesce` reads as an hour that is not on a European Union leg.

Specific fuel consumption for the main engine is corrected for load, because a diesel engine is least efficient at very low and very high load. The correction is equation (10) of the Fourth IMO Greenhouse Gas Study, reproduced here.

$$SFC_{ME,i} = SFC_{base}\left(0.455\,Load_i^2 - 0.710\,Load_i + 1.280\right) \tag{10}$$

$SFC_{ME,i}$ is the load-corrected specific fuel consumption of the main engine at hour i in grams per kilowatt hour, SFC_{base} the baseline value for the engine type, fuel, and build year, and $Load_i$ the load factor from equation (7). The parabola is minimised near 80% load, the design operating point of a marine diesel engine. Auxiliary and boiler values are not load corrected, following the same source, and the low-load adjustment factor is not applied to CO_2 because Table 20 of that study assigns CO_2 a factor of 1.00 at every load.

Hourly fuel consumption sums the three consumers, each at its own specific consumption, and annual emissions follow by applying the carbon factor of the assigned fuel and summing over the year, as shown in equations (11) and (12).

$$FC_i = \left(W_{ME,i}\,SFC_{ME,i} + W_{AE,i}\,SFC_{AE} + W_{BO,i}\,SFC_{BO}\right)\Delta t \tag{11}$$

$$E_{ship,y} = \sum_{i \in y} \frac{FC_i\,EF_f}{10^6} \tag{12}$$

FC_i is the fuel consumed in hour i in grams, $W_{ME,i}$, $W_{AE,i}$, and $W_{BO,i}$ the three power demands in kilowatts, SFC_{AE} and SFC_{BO} the uncorrected auxiliary and boiler consumption rates, Δt the time step of one hour, EF_f the carbon conversion factor of the fuel assigned to that hour in grams of CO_2 per gram of fuel, $E_{ship,y}$ the annual total in tonnes, and the divisor the conversion from grams to tonnes. The factors are those of Table 21 of the Fourth IMO Greenhouse Gas Study, whereas Selin et al. (2021) used the corresponding factors of the Third Study (IMO, 2015), current at the time of their work.

Table 7 gives the values these equations require for the two pilot vessels. The complete tables, covering every engine type, fuel and build-year band, are reproduced in Appendix C as Tables C7 to C9.

Table 7. Specific fuel consumption and CO_2 emission factors.

Constant	HFO	MDO	Source
Main engine baseline SFC, slow-speed diesel built after 2001 (g kWh^{-1})	175	165	IMO (2020), Table 19
Auxiliary engine SFC (g kWh^{-1})	195	185	IMO (2020), Table 19
Auxiliary boiler SFC (g kWh^{-1})	340	320	IMO (2020), Table 19
CO_2 conversion factor (g CO_2 per g fuel)	3.114	3.206	IMO (2020), Table 21
Load correction coefficients	0.455, -0.710, 1.280	0.455, -0.710, 1.280	IMO (2020), equation (10)
Low-load adjustment factor for CO_2	1.00	1.00	IMO (2020), Table 20
Minimum main-engine load	0.07	0.07	IMO (2020)

3.6. Allocate emissions to countries

The quantity to be allocated is the CO_2 attributable to international voyages, not the total of the ship, so the annual figure must first be split:

- Hours on labelled legs are assigned directly to the corresponding domestic or international category.
- Unlabelled hours, such as those before the first leg, after the last leg, or missing port country data, are divided between domestic and international categories based on the proportion of directly labelled emissions for that vessel and year.
- Emissions, not hours, determine this proportion because differing port and sea emission rates would otherwise bias the split.
- Each vessel's full international CO_2 total is attributed to a single country under each allocation option.

Four of the five allocation options are computed: flag, registered owner, ISM manager, and operator country. Equasis has no operator field, so we applied the commercial manager as a proxy. The roles are related but not identical, and where one company holds both the operator and manager options collapse into each other, reducing the contrast the original study reported. Additionally, Equasis reports company addresses not the countries of incorporation, so the country key is inferred from an address.

The fifth option of Selin et al. (2021), allocation to the country where bunker fuel is sold, is not computed. National bunker sales statistics report fuel delivered within a country and cannot identify where an individual ship took fuel, so the option is a fleet scale construct with no meaning for two vessels. A fleet extension would recover it from the international marine bunker flows in the United Nations energy statistics database (United Nations Statistics Division, 2025), converting reported tonnes of residual and distillate fuel with the factors of Table 7. Finally, territory alignment maps Hong Kong to China and the Isle of Man to the United Kingdom, following the supplementary treatment of Selin et al. (2021).

3.7. Compute allocation impacts

Selin et al. (2021) used the 2018 edition of the Global Carbon Budget, and we use the 2025 edition. The workbook publishes megatonnes of carbon, so the baseline is multiplied by 3.664, the ratio of the molecular mass of carbon dioxide to the atomic mass of carbon. A country with no baseline raises an error, since a zero would silently produce an infinite percentage increase. The relative impact follows in equation (13), and the absolute impact is simply the allocated total expressed in megatonnes.

$$\Delta E\%_c = \frac{100\, E_{c,o}}{10^6 \times 3.664\, B_c} \tag{13}$$

$\Delta E\%_c$ is the increment to the carbon budget of country c as a percentage of its national baseline, $E_{c,o}$ the allocated total in tonnes, B_c the baseline in megatonnes of carbon, and the divisor converts tonnes of CO_2 to megatonnes and the baseline from carbon to CO_2. Pairing an absolute with a relative measure follows Selin et al. (2021), whose headline results are expressed as percentage increases to national carbon budgets.

Ranks and concentration shares are computed because the purpose is to demonstrate a pathway that scales, but they carry no substantive meaning for two ships and are not interpreted here.

4. Results and Discussion

The pipeline was implemented end to end for two vessels between 1 January 2017 and 31 December 2024, a span of 70,128 hours each. Table 8 lists the pair with the fields Equasis supplied and the activity counts the events endpoint returned over those eight years. COSCO ITALY is a large container ship whose four allocation keys all resolve to one country; it concentrates 389 calls in 17 countries. RCC AMERICA is a smaller vehicle carrier on an open registry, and it spreads 588 calls across 64 countries. Every vessel-specific value, country key, and model switch is held in configuration, so extending the work is a loop over IMO numbers. What follows reports what the demonstration establishes about the procedure itself, where the open substitutions hold, where they cost accuracy, and what a fleet-scale extension would have to add.

Table 8. Configured pilot vessels used for testing the pipeline.

IMO	Ship	Label (in code)	Flag	Type	DWT (t)	LOA (m)	Names used by presence pull	Port calls	Port countries	Intl. legs
9516454	COSCO ITALY	A	HKG	container	156,610	365.9	COSCO ITALY	389	17	225
9277802	RCC AMERICA	B	BHS	vehicle	21,182	199.8	RCC AMERICA, HOEGH AMERICA	588	64	395

4.1. Reconstructing ship activity from an open access AIS product

Figure 4 presents the activity coverage retrieved from the GFW database. The raw measure is the share of all calendar hours in a year for which the presence product returned a position. The active measure is the share of in-service hours, which are the calendar hours remaining once the intervals classified as out of service in Section 3.1 are removed. This measure is the relevant one because an out-of-service hour is not a missing observation. For instance, COSCO ITALY has a 6,772-hour gap in the GFW presence data from 10 June 2019 to 18 March 2020. This yields 36.1% raw coverage in 2019 and a naïve 2.77× correction. Section 3.1 assigns to it no emissions at all, so nothing is lost by its absence, whereas a reception gap inside a trading period is filled by interpolation and does carry modelled emissions.

Nonetheless, active coverage is incomplete in several ship-years, concentrated in time rather than scattered. Interpolation supplies 11.9% and 13.1% of modelled hours between 2017 and 2021 against 1.1% and 2.0% between 2022 and 2024, and the shift appears on both hulls at once, which apparently points to reception of AIS signals, and not to the behavior of either ship. The consequence is that the early years rest more heavily on inferred speeds than the late ones, and because speed enters the power calculation cubed, they are also the years in which the smoothing choice examined below does the most work. A study window reaching further back than 2017 should expect that dependence to grow.

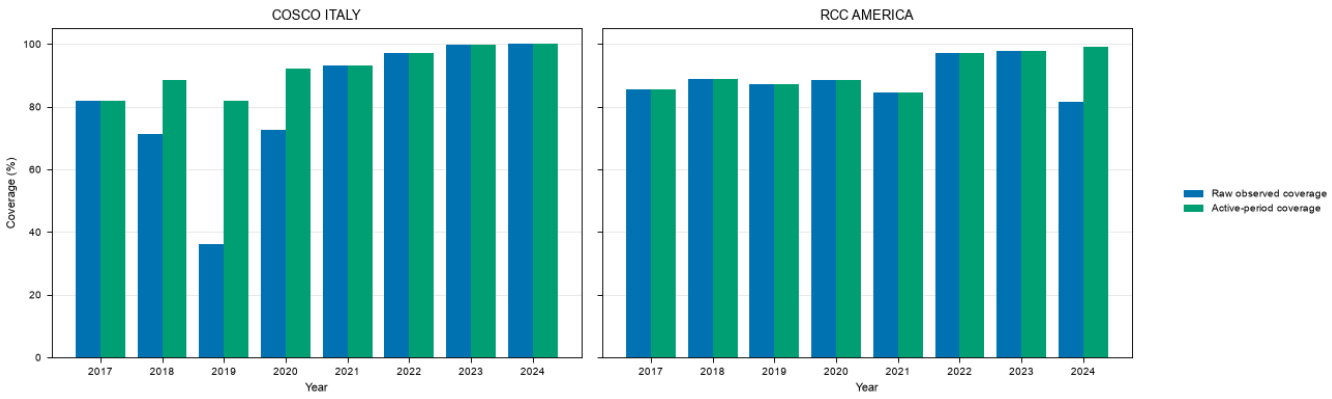


Figure 4. Annual AIS coverage for the two pilot vessels, as a share of all calendar hours and as a share of in-service hours.

The smoothing treatment of speed is one of the largest differences in the methodology versus the original paper, whose records carried a transmitted speed. The eight-year emissions total falls by 19.3% and 7.9% between the raw and the seven-hour series. That makes the smoothing window a larger source of variation than the treatment of missing hours, which moves the same totals by only 1.6% and 1.3%. However, it is not a defect specifically of open data. Psaraftis and Lagouvardou (2023) note that transmitted AIS speed is itself a SOG, so every AIS-driven power model already carries an approximation for tides and currents, and cell quantization adds a second layer to it that smoothing reduces but does not remove.

Table 9 collects the diagnostics behind these statements, running from the raw hour counts through the activity, fuel, and voyage classifications to the eight-year emission ranges. The container ship spends more of its life inside a port visit and less inside an emission control area, while the vehicle carrier calls at nearly four times as many countries, sails a higher proportion of international legs, and burns distillate for more than twice the share of its hours.

Table 9. Activity and emission diagnostics for the two pilot vessels, 2017 to 2024.

Diagnostic	COSCO ITALY (container)	RCC AMERICA (vehicle)
Observed presence hours	57,151	62,322
Linearly infilled hours	4,497	6,255
Out-of-service hours excluded	8,480	1,551
Active coverage, range by year	82.0 to 100.0%	84.5 to 99.1%
Port calls, all at confidence 4	389	588
Hours inside a recorded port visit	17,427	15,450
Hours inside an emission control area	4,310	8,489
Cubic speed bias, w = 1 to w = 7	1.94 to 1.26	2.06 to 1.34
Eight-year CO_2, range across estimates (t)	473,408 to 901,548	182,511 to 191,745
International share of CO_2, 2024	97.5%	93.6%

Source: pipeline checkpoints. Cubic speed bias is the ratio of the mean cube of speed to the cube of the mean speed. CO_2 ranges are taken at a three-hour smoothing window.

4.2. Ship specifications as the dominant source of uncertainty

Table 10 collects what the estimation routes of Section 3.3 produce for each ship. Each row carries the reference speed, the installed power, the reference load fraction at which that speed is attained, and the eight-year CO_2 total the combination generates. For COSCO ITALY the three routes span a factor of 1.90, from 473,408 to 901,548 tonnes. For RCC AMERICA the two available routes span 1.05.

Table 10. Power and reference-speed estimates for the two pilot vessels, and the eight-year CO_2 each produces.

Vessel	Estimate	V_{ref} (kn)	MCR (kW)	f_{ref}	Total CO_2 (t)	Ratio
COSCO ITALY	A, EEXI curve fit	25.55	67,912	0.75	473,408	1.00
	B, Admiralty, calibrated	22.70	79,332	1.00	901,548	1.90
	D, EPA DWT power with Froude speed	22.70	92,868	0.83	900,528	1.90
RCC AMERICA	A, EEXI curve fit	19.96	14,086	0.75	191,745	1.05
	D, EPA DWT power with EEXI speed	19.96	13,282	0.75	182,511	1.00

That spread does not follow installed power. Estimate B carries 17% more installed power than Estimate A yet produces 90% more CO_2, and Estimate D carries 37% more power than A yet lands within 1% of B. The reference load fraction resolves the apparent inconsistency, because effective propulsion at a given observed speed varies as installed power multiplied by that fraction and divided by the cube of the reference speed, and on that basis Estimates B and D imply about 2.2 times the propulsion of Estimate A. The CO_2 totals compress that ratio to 1.90 and 1.90, since auxiliary and boiler demand does not scale with propulsion and roughly a quarter of the period is spent alongside.

The vehicle carrier's methods share a reference speed and a reference load fraction, so their difference reflects only the disagreement between two power regressions. The container hull is where speed is genuinely in dispute, and there one estimate can already be set aside, since Estimate A places the reference speed at 25.55 knots, above the container fleet envelope of 24.5 knots (Cepowski & Chorab, 2021).

The estimates that remain are approximations in their own right, and Tu et al. (2018) show as much for the Admiralty coefficient on container speed and power curves, proposing hull-form coefficients in place of its constant exponent. The wider literature indicates the scale of what is at stake. Sun et al. (2026) found that the imputation method chosen for missing ship technical parameters moves estimated port CO_2 by 2% to 6%, and that poor data quality can understate emissions by at least 12%, in datasets where the parameter is absent for only a fraction of ships, whereas here it is absent for every ship. Merien-Paul et al. (2018), comparing bottom-up estimates against in-situ measurement on a capesize bulk carrier, found a deadweight-based installed power of 18,252 kilowatts against an actual 16,700, and assumed seagoing load factors of 0.7 to 0.8 against a measured 0.51. Thus, the limiting factor for this method is not the model but a single register field, and any extension of this work should acquire installed power and reference speed first.

Neither of the models summarized in Section 2.1 was adopted, for practical reasons. The resistance approach of Johansson et al. (2017) needs, beyond the hull-specific speed-power curve and the instantaneous draught already noted, a block coefficient, a wetted surface and a propeller diameter for every vessel, so it could only be run here at an assumed hull form and an assumed loading condition, which would forfeit the physical fidelity that motivates

it. The regression cascade of Schwarzkopf et al. (2021) is affordable, requiring only the ship type and the gross tonnage, but it is calibrated on a fleet in which small vessels dominate and it places container ships in the same class as general cargo ships, so its square-root relationship returns approximately 25,000 kilowatts even at a gross tonnage of 200,000, roughly a third of the power installed in a container ship of that size, and its service-speed function takes that same estimated power as its argument, which leaves the reference speed and the installed power derived from a single quantity and unable to constrain each other. Its load equation is in any case the cubic law of equation (7), so it offers no alternative treatment of the relationship between speed and power.

For investigations supported by registries that provide detailed hull geometry and environmental data, the resistance-based approach offers superior precision. This level of physical fidelity is essential when analyzing fleets with significant ballast trading, as load-factor models fail to differentiate between laden and ballast conditions at equivalent speeds. Conversely, for regional inventories encompassing small or unregistered vessels lacking formal records, the regression-based framework of Schwarzkopf et al. (2021) represents the most viable alternative. However, utilizing such proxies involves a trade-off in accuracy; their reported prediction intervals, reflecting a potential 62% increase or 14% decrease in estimated CO_2 at the 95% confidence level, quantify the inherent uncertainty of substituting estimated for observed main-engine power.

4.3. behavior of the emission model

COSCO ITALY spends 24.8% of its in-service hours at berth and 68.9% in slow transit, yielding 4.7% and 87.8% of its CO_2, while RCC AMERICA spends 17.9%, 61.0% and 12.3% of its hours at berth, in slow transit and in normal cruising, for 5.4%, 62.5% and 29.8% of its CO_2. Of the 140,256 vessel-hours, 9.1% fall inside an emission control area and 4.6% on a leg between two European Union ports, which puts distillate on 7.5% of the container ship's hours and 16.4% of the vehicle carrier's, and the berth rule of Section 3.4 keeps the port hours on the right auxiliary load, since the distance test alone placed 1,143 hours at berth against 17,427 hours inside recorded port visits.

In Figure 5 we can observe that auxiliary engine and boiler fuel is identical under every estimate because Table 17 reads demand from ship type, size band and operating mode and never from installed power. Hence, the whole disagreement between estimates sits in the main engine bar, which rises from 125,583 tonnes under Estimate A to 262,803 and 262,487 tonnes under B and D. Improving the specification input is the only route to a narrower answer.

Figure 6 resolves the same totals by year. The ratio between estimates is near-constant, lying between 1.84 and 1.92 in every one of the container ship's eight years and between 1.048 and 1.052 in the vehicle carrier's, so the choice of power estimate rescales a series, but does not change the behavior. Quantities expressed as ratios are correspondingly more robust to the specification gap than absolute tonnages are. That robustness does not transfer automatically to an allocation, because the rescaling factor differs by hull, 1.90 on the container ship against 1.05 on the vehicle carrier, so a country whose allocated fleet is dominated by one ship type would move relative to a country dominated by another.

The one exposure no scenario axis bounds is the set of load corrections Section 3.4.1 could not apply. Bialystocki and Konovessis (2016) indicate its size, having fitted 418 noon reports from a pure car and truck carrier, the same type as RCC AMERICA, to separate the effects of draught and displacement, weather force and direction, and hull and propeller roughness on the fuel consumption and speed curve. Those terms shift the curve enough to matter for voyage planning, so the figures reported here carry a bias whose sign depends on the loading and weather history of each hull and which neither the smoothing nor the specification axis captures.

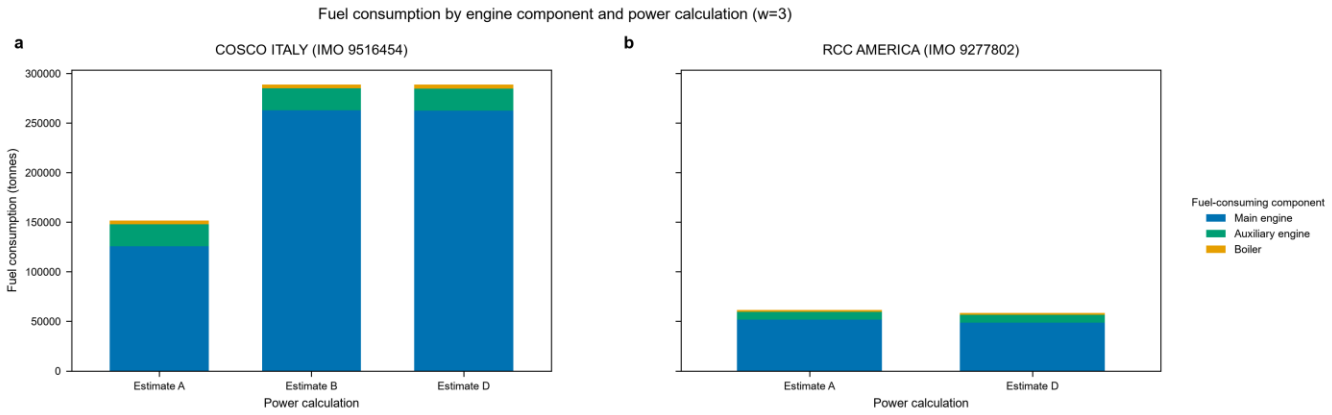


Figure 5. Fuel consumption by engine component and power estimate, 2017 to 2024, at a three-hour smoothing window.

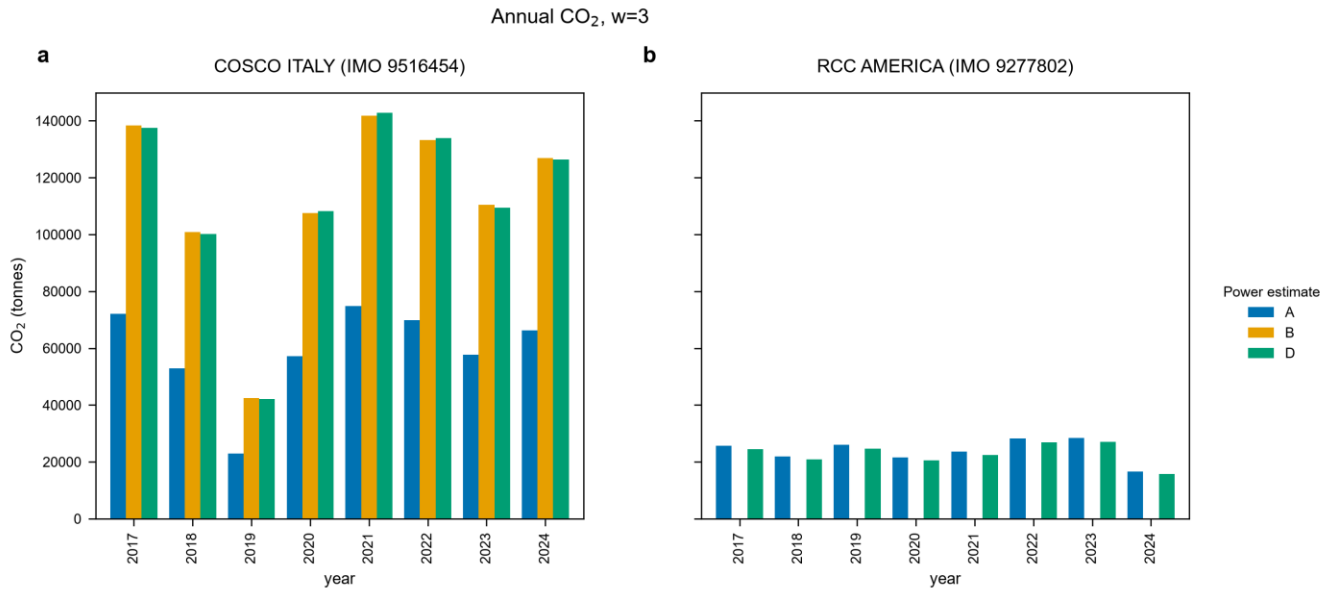


Figure 6. Annual CO_2 by power estimate, at a three-hour smoothing window.

4.4. Differences from Selin et al. (2021)

The choices in this methodology do not all cost the same in terms of accuracy of the model, and Table 11 orders them by the effect measured here. The specification substitution, which replaces one commercial register field with three published regressions, is worth a factor of 1.90. The activity substitution, which replaces transmitted AIS positions and speeds with an aggregated presence product, is worth 19.3% through the speed derivation alone. The gap treatment, which looks like the most consequential change because it alters what an annual total means, is worth less than 2%. The last two rows change no CO_2 quantity at all, since the fuel and berth refinements make the classification more faithful without moving the total appreciably, and the newer carbon budget alters only the denominator of the percentage impacts.

Table 11. Substitutions made in this reconstruction and their measured effect.

Element	Substitution	Effect measured here
Ship specifications	Three published estimation routes replace a commercial register field	Factor of 1.90 on the container hull
Activity data	Aggregated hourly presence replaces transmitted AIS positions and speeds	19.3% across the smoothing window, with a residual bias smoothing reduces but does not remove
Gap treatment	In-service hours are infilled and out-of-service hours excluded rather than scaled	1.6% and 1.3%, and an annual total becomes an observed rather than a grossed-up figure
Fuel and berth rules	Observed legs and port visit intervals replace activity ranges and distance thresholds	No material change to CO_2, and a more faithful classification of some 16,000 hours
Carbon budget and scope	The 2025 budget edition, two vessels, and four allocation options	No allocated quantity changes, no distributional result, and no bunker option

4.5. Requirements for scaling, and principal limitations

The pipeline has been built to be able to scale this methodology to more ships and countries. However, there are gaps in the data acquisition that restrict the automatic pull of more GFW datapoints. The emissions and allocation stages already loop over IMO numbers without modification, the checkpoints are written per vessel, and nothing in the model couples one hull to another, so the computation is parallel by construction.

The acquisition steps that do not scale as they stand are:

- Vessel discovery cannot rely on the ship-name filter used here, so a fleet pull must sweep flag and vessel type at world extent and select on the IMO number carried in each returned record, raising the volume retrieved by roughly two orders of magnitude and making request quota the binding constraint.
- Registry acquisition remains a significant obstacle because Equasis and Marine Traffic expose no programmatic interface, require an authenticated session, and owner-country selection criteria could not be automated, necessitating a licensed register or a systematic imputation layer.
- The bunker option requires no new engineering beyond expanding to fleet scale, as Section 3.6 already identifies the necessary statistics.
- Estimate C remains unpopulated because no sourced specifications could be obtained for either hull, leaving reference figures as ranges instead of single values. Future researchers should evaluate which method provides more accurate results based on the assumptions stated here and the data they have access to.

5. Conclusions

This paper set out to establish whether the national allocation of shipping CO_2 can be reconstructed from openly accessible data, and the two-vessel demonstration indicates that it can, subject to the modifications documented in Section 4.4. In our model, emissions are resolved to the vessel-year and split into domestic and international components before any country key is applied, so the four actor-based options computed here are interchangeable with any other rule that attaches to a voyage or a vessel, including the trade-linked and network-based frameworks of Fry et al. (2024) and Zhao et al. (2025), without rebuilding the emission model. Because every vessel-specific value and model switch is held in configuration, extending the study is a loop over IMO numbers, and because the choice of power estimate rescales an annual series almost uniformly, the concentration shares and rank orderings on which allocation debates actually turn are considerably more robust than the absolute tonnages that carry them.

A method that any researcher can run, audit, and re-run against a current carbon budget edition allows the distributional evidence underpinning the responsibility allocation debate to be independently verified and updated.

Improving the accuracy of what the procedure returns is a data acquisition problem before it is a modeling one. The dominant uncertainty is the ship specification, which spans a factor of 1.90 in the eight-year total of the container hull, against less than 2% for the treatment of missing hours, so the single most valuable addition is a licensed register supplying installed power and reference speed as observed fields. Moreover, some refinements would narrow the remaining bands. Position-level AIS carrying transmitted speed would remove the cell quantization that smoothing reduces but does not eliminate, and which alone moves the totals by 19.3%. A draught, weather, and hull condition correction of the kind fitted by Bialystocki and Konovessis (2016) would bound the one exposure that neither the smoothing nor the specification axis currently captures. With those changes in place, the procedure would support the distributional claims that the present analysis deliberately withholds.

Code and Data Availability

Analysis, plotting, and processing scripts are available here: https://github.com/arantzazu-arregui/emissions-allocation

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Appendix A. List of Abbreviations

AIS	Automatic Identification System
CBDR	Common but differentiated responsibilities and respective capabilities
DWT	Deadweight tonnage. This is the total carrying capacity of the ship including cargo, fuel, crew, passengers, and food
ECA	Emission control area
EEXI	Energy Efficiency Existing Ship Index
EEZ	Exclusive economic zone
EF	Emission factor
EMSA	European Maritime Safety Agency
EU	European Union
GCB	Global Carbon Budget
GFW	Global Fishing Watch
GHG	Greenhouse gas
GT	Gross tonnage. A measure of a ship's total internal volume (tonnes)
HFO	Heavy fuel oil
IEA	International Energy Agency
IHS	IHS Maritime and Trade, publisher of the World Register of Shipping
IMO	International Maritime Organization
ISM	International Safety Management Code, and by extension the manager certified under it
LBP	Length between perpendiculars (m). This is the distance between the rudder post, and the point where the bow meets the waterline
LOA	Length overall.
Load	The fractional portion of a main engine's total installed power that is being used
MARPOL	International Convention for the Prevention of Pollution from Ships
MCR	Maximum continuous rating of the main engine
MDO	Marine diesel oil
MEPC	Marine Environment Protection Committee of the IMO
MGO	Marine gas oil
MMSI	Maritime Mobile Service Identity
MRV	Monitoring, reporting and verification
OECD	Organisation for Economic Co-operation and Development
SBSTA	Subsidiary Body for Scientific and Technological Advice of the UNFCCC
SFC	Specific fuel consumption
SOG	Speed over ground
SSD	Slow-speed diesel
TEU	Twenty-foot equivalent unit
UNCTAD	United Nations Conference on Trade and Development
UNFCCC	United Nations Framework Convention on Climate Change
US EPA	United States Environmental Protection Agency
Vref	Reference speed of the ship. Service speed: The intended operating speed of the ship; Maximum speed: The maximum speed that the ship can achieve in calm waters

WRS	World Register of Shipping
YAML	YAML Ain't Markup Language, the configuration format used by the pipeline

Other definitions:

Operating Mode: The type of activity the ship is engaged in:

- At Berth: not moving at a berth in a port
- Anchored: floating at anchor
- Maneuvering: moving at low speed within a port area
- Transit: traveling in open water

Appendix B. Data preprocessing equations and constants

This appendix collects the equations and constants that prepare the activity data for the physical model of Section 3. They convert raw positions into speeds, describe how completeness is measured, and record the two auxiliary conversions used to classify a ship and to index it against the tabulated auxiliary loads. None of them estimates power or emissions.

The great-circle distance between two consecutive observed positions is computed with the haversine formulation of equations (B1) and (B2), and the derived speed over ground follows in equation (B3).

$$a = sin^2\left(\frac{\Delta\varphi}{2}\right) + cos\varphi_1 cos\varphi_2 sin^2\left(\frac{\Delta\lambda}{2}\right) \quad \text{(B1)}$$

$$d = 2Rarcsin\left(\sqrt{a}\right) \quad \text{(B2)}$$

$$SOG_{raw,i} = \frac{d_i}{1.852\,\Delta t_i} \quad \text{(B3)}$$

a is the dimensionless haversine of the central angle between the two positions, $\Delta\varphi$ and $\Delta\lambda$ are the differences in latitude and longitude in radians, φ_1 and φ_2 are the latitudes of the earlier and later positions in radians, d is the great-circle distance in kilometres, R is the mean earth radius of 6,371.0088 km, $SOG_{raw,i}$ is the derived speed at hour i in knots, d_i is the distance to the previous non-missing observation, Δt_i is the elapsed time between them in hours, and 1.852 is the number of kilometres in a nautical mile. Equations (B1) and (B2) are the standard haversine construction and are not specific to this study. Selin et al. (2021) required no equivalent step, because their AIS records carry the speed transmitted by the ship.

Completeness of the activity record is reported in the two forms of equation (B4).

$$coverage_{raw} = \frac{h_{obs}}{h_{elapsed}}, \qquad coverage_{active} = \frac{h_{obs}}{h_{elapsed} - h_{inactive}} \quad \text{(B4)}$$

h_{obs} is the number of observed vessel-hours in the year, $h_{elapsed}$ the number of hours in the calendar year, and $h_{inactive}$ the number classified as out of service. The active form is the one used to flag a vessel-year as low confidence, since hours in which the ship was laid up should not count against observation quality.

The vessel-level domestic test of Selin et al. (2021), retained here as a diagnostic rather than as the allocation scope, is given in equation (B5).

$$domestic \quad if \quad \max_{c} \left(\frac{h_c}{H_{active}} \right) > 0.95 \tag{B5}$$

h_c is the number of active vessel-hours whose position falls inside the exclusive economic zone of sovereign state c, H_{active} is the total number of active vessel-hours in the year including hours on the high seas, and the maximum is taken over all sovereign states. Hours matching no zone are treated as high seas and remain in the denominator.

Container capacity in twenty-foot equivalent units, which Equasis does not publish but the auxiliary load tables require, is estimated from the moulded beam by the preliminary-design formula of Cepowski and Chorab (2021), given in equation (B6).

$$TEU = \left(\frac{B_m}{3.27} \right)^{1/0.29} \tag{B6}$$

TEU is the container capacity in twenty-foot equivalent units, B_m is the moulded beam in metres, and 3.27 and 0.29 are the fitted coefficients. Ship types indexed by deadweight tonnage, including the vehicle carrier, are matched to their size band directly and need no conversion.

Table B1. Constants and thresholds used in activity ingestion and gap treatment.

Constant	Value	Role	Source
Mean earth radius, R	6,371.0088 km	Equation (B2)	Standard geodetic constant
Nautical mile	1.852 km	Equation (B3)	International standard
Spatial resolution	HIGH, 0.01 degree	Cell size of the presence grid	Global Fishing Watch (2025)
Temporal resolution	HOURLY	One record per vessel-hour	Global Fishing Watch (2025)
Smoothing windows, w	1, 3, 5, 7 hours	Equation (1)	This study, set by the observed cubic bias
Minimum annual coverage	10 per cent of elapsed hours	Rejects a partially returned annual pull	This study
Maximum annual coverage	101 per cent of elapsed hours	Rejects a duplicated annual pull	This study
Low-confidence coverage flag	below 95 per cent	Flags a vessel-year without rejecting it	This study
Out-of-service gap threshold	7 days with no port call	Separates inactivity from missing observation	This study
Domestic classification threshold	0.95	Equation (B5)	Selin et al. (2021)
Container capacity coefficients	3.27 and 0.29	Equation (B6)	Cepowski and Chorab (2021)
Carbon to carbon dioxide ratio	3.664	Baseline conversion in equation (13)	Stoichiometric constant
Container fleet speed envelope	6.0 to 24.5 knots	Plausibility check on a reference speed	Cepowski and Chorab (2021), Table 1
Horsepower to kilowatt conversion	0.745699872	Equation (6)	International standard

Appendix C. Complete published parameter sets

Section 3 reports only the parameter values for the selected two ships for testing. This appendix reproduces the published sets in full, for every ship type, engine type, fuel and size band that the sources cover, so that a researcher extending the work to a larger fleet has the whole lookup in one place and does not have to re-transcribe it from the source documents. Each table names the document and, where applicable, the printed table number it comes from. The tables are held in the pipeline as configuration data rather than as code.

Table C1. EEXI average reference speed and installed power coefficients, all twelve ship categories.

Ship category	Capacity basis	A	C	Speed cap	D	F	Power cap
Bulk carrier	DWT	10.6585	0.02706	none	23.7510	0.54087	none
Gas carrier	DWT	7.4462	0.07604	none	21.4704	0.59522	none
Tanker	DWT	8.1358	0.05383	none	22.8415	0.55826	none
Container ship	DWT	3.2395	0.18294	80,000	0.5042	1.03046	95,000
General cargo	DWT	2.4538	0.18832	none	0.8816	0.92050	none
Refrigerated bulk	DWT	1.0600	0.31518	none	0.0272	1.38634	none
Combination carrier	DWT	8.1391	0.05378	none	22.8536	0.55820	none
LNG carrier	DWT	11.0536	0.05030	none	20.7096	0.63477	none
Vehicle carrier	DWT	16.6773	0.01802	none	262.7693	0.39973	none
Ro-ro	DWT	8.0793	0.09123	none	37.7708	0.63450	none
Ferry, ro-pax	DWT	4.1140	0.19863	none	9.1338	0.91116	none
Cruise	GT	5.1240	0.12714	none	1.3550	0.88664	none

Source: International Maritime Organization (2021), Resolution MEPC.333(76), Appendix. Reference speed is A times capacity raised to C, in knots, and installed power is D times capacity raised to F, in kilowatts. The capacity caps apply to container ships only; above the cap the estimate is flat rather than extrapolated. Cruise ships with non-conventional propulsion take gross tonnage as the capacity basis, not deadweight. A separate capacity correction applies to vehicle carriers whose ratio of deadweight to gross tonnage falls below 0.35, multiplying capacity by that ratio divided by 0.35 raised to the power of minus 0.8. Neither pilot vessel triggers it.

Table C2. Regressions of rated main engine horsepower on deadweight tonnage, all ship types.

Ship type	Intercept (hp)	Slope (hp per DWT)	N	R^2	Table
All ships	9,070	0.1097	4,103	0.14	4-3
Auto carrier	7,602	0.4172	157	0.176	4-3
Bulk carrier	6,726	0.0985	1,644	0.55	4-3
Container	-749.4	0.800	489	0.59	4-3
General cargo	3,046	0.288	641	0.56	4-3
Passenger	-4,877	6.81	40	0.72	4-3
Reefer	1,364	1.007	160	0.58	4-3
Ro-ro	4,358	0.5364	110	0.76	4-3
Tanker	6,579	0.1083	861	0.66	4-3
Bulk carriers and tankers	9,070	0.101	2,505	0.67	4-5
Passenger	-4,877	6.81	40	0.72	4-5
General cargo	3,046	0.288	641	0.56	4-5

Container, ro-ro, auto carrier and reefer	2,581	0.719	917	0.71	4-5

Source: United States Environmental Protection Agency (2000), EPA420-R-00-002, Table 4-3 (type-specific fits) and Table 4-5 (recommended consolidated categories). Horsepower is converted to kilowatts by a factor of 0.745699872. The two relations used in this study, the container fit and the consolidated container, ro-ro, auto carrier and reefer fit, were verified against the report. The remaining rows are reproduced for completeness and should be checked against the source before being relied on. Table 4-5 exists because several of the type-specific fits in Table 4-3 are statistically weak, the auto carrier in particular, so the consolidated relation is the one the report recommends.

Table C3. Values used in the calculation of Estimate B for the container pilot.

Quantity	Symbol	Value	Source or basis
Length between perpendiculars	LBP	347.6 m	Estimated as 0.95 times the length overall of 365.90 m
Moulded beam	B	51.20 m	Public vessel registers, observed
Draught	T	16.0 m	Public vessel registers, observed
Deadweight	DWT	156,610 t	Equasis, observed
Froude number range	Fn	0.19 to 0.21	Cepowski and Chorab (2021), modern container fleet mean 0.21
Block coefficient	CB	0.61	Charchalis (2014), assigned for container hulls
Seawater density	ρ	1.025 t m-3	Standard constant
Deadweight to displacement ratio		0.80	Charchalis (2014)
Admiralty coefficient	Cadm	482, range 352 to 593	Back-calculated in Table C4 from Charchalis (2014), Table 1
Reference speed, equation (4)	Vref	21.57-23.84 kn, midpoint 22.70	Computed from Fn and LBP
Displacement, geometric	Δ	178,000 t	CB times LBP times B times T times ρ
Displacement, deadweight ratio	Δ	195,800 t	DWT divided by 0.80
Displacement, point scenario	Δ	186,900 t	Mean of the two, which differ by 10.0%
Installed power, equation (5)	MCR	79,332 kW	Point scenario, midpoint speed and mean displacement
Installed power, endpoint range	MCR	65,900-94,700 kW	Froude and displacement endpoints combined
Installed power, coefficient range	MCR	64,500 to 108,600 kW	Cadm at 593 and 352, point scenario otherwise

Source: hull dimensions from public vessel registers and Equasis, constants as attributed. The length between perpendiculars is not published for this hull and is corroborated at 344.4 m by the deadweight relation of Cepowski and Chorab (2021). The final row is a diagnostic on the coefficient alone and is not carried as a separate scenario. No published Admiralty coefficient, block coefficient or Froude range is available for vehicle carriers, ro-ros, bulk carriers, tankers, general cargo ships, gas carriers, refrigerated bulk carriers, ferries or cruise ships, so the pipeline raises an error for any such type rather than borrowing the container values.

Table C4. Back-calculation of the Admiralty coefficient from the comparison ships of Charchalis (2014), Table 1.

Ship	TEU	Design speed (kn)	Installed power (kW)	Displacement (t)	Cadm
1	1,334	19	12,200	27,072	506.9
2	1,253	18	10,640	28,495	511.4
3	1,388	17	9,540	24,280	431.8
4	1,346	18	11,300	32,570	526.4
5	1,228	19	13,560	22,215	399.7
6	1,300	19	11,130	25,000	526.9
7	1,300	21	17,760	25,054	446.5
8	1,388	19	12,180	25,690	490.3
9	1,216	19	12,180	23,370	460.3
10	1,334	19	12,180	27,072	507.7
11	1,202	18	11,300	38,930	592.8
12	1,338	17	11,130	42,157	534.7
13	1,388	18	11,130	27,200	473.9
14	1,174	19	13,320	24,540	434.9
15	1,384	20	16,200	30,453	481.6
16	1,262	18	13,530	23,320	351.8
17	1,388	17	9,540	25,723	448.8

Source: columns two to five are reproduced from Charchalis (2014), Table 1. That source reports no Admiralty coefficient and does not state the relationship, so the final column is computed for this study as the displacement raised to two thirds, multiplied by the cube of the design speed, divided by the installed power, which is equation (5) rearranged to solve for the coefficient rather than for the power. Displacement is in tonnes, speed in knots and power in kilowatts, and because the coefficient is dimensional its numerical value holds only for that combination of units. Across the seventeen ships the median is 482, the mean 478, the standard deviation 57 and the range 352 to 593, and the median is the value adopted in Table C3. Since installed power varies inversely with the coefficient, that range maps to 1.37 down to 0.81 times the point estimate. Ships 11 and 12 carry displacements of 38,930 and 42,157 tonnes against a sample median near 25,700 tonnes at comparable capacities, and ship 12 implies a deadweight to displacement ratio of 0.478, so both rows are treated as probable transcription errors in the source. Excluding them moves the median to 474 and the range to 352 to 527, which leaves the point estimate essentially unchanged and narrows the upper bound.

Table C5. Auxiliary engine and auxiliary boiler power output in kilowatts, all ship types and size bands.

Ship type	Size unit	Size band	AE berth	AE anchor	AE manoeuvre	AE sea	AB berth	AB anchor	AB manoeuvre	AB sea
Container	TEU	up to 999	370	450	790	410	250	250	240	0
		1,000 to 1,999	820	910	1,750	900	340	340	310	0
		2,000 to 2,999	610	910	1,900	920	460	450	430	0
		3,000 to 4,999	1,100	1,350	2,500	1,400	480	480	430	0
		5,000 to 7,999	1,100	1,400	2,800	1,450	590	580	550	0

		8,000 to 11,999	1,150	1,600	2,900	1,800	620	620	540	0
		12,000 to 14,499	1,300	1,800	3,250	2,050	630	630	630	0
		14,500 to 19,999	1,400	1,950	3,600	2,300	630	630	630	0
		20,000 and above	1,400	1,950	3,600	2,300	700	700	700	0
Bulk carrier	DWT	up to 9,999	110	180	500	190	70	70	60	0
		10,000 to 34,999	110	180	500	190	70	70	60	0
		35,000 to 59,999	150	250	680	260	130	130	120	0
		60,000 to 99,999	240	400	1,100	410	260	260	240	0
		100,000 to 199,999	240	400	1,100	410	260	260	240	0
		200,000 and above	240	400	1,100	410	260	260	240	0
Oil tanker	DWT	up to 4,999	250	250	375	250	500	100	100	0
		5,000 to 9,999	375	375	560	375	750	150	150	0
		10,000 to 19,999	690	500	580	490	1,250	250	250	0
		20,000 to 59,999	720	520	600	510	2,700	270	270	270
		60,000 to 79,999	620	490	770	560	3,250	360	360	280
		80,000 to 119,999	800	640	910	690	4,000	400	400	280
		120,000 to 199,999	2,500	770	1,300	860	6,500	500	500	300
		200,000 and above	2,500	770	1,300	860	7,000	600	600	300

Chemical tanker	DWT	up to 4,999	110	170	190	200	670	160	130	0
		5,000 to 9,999	330	490	560	580	670	160	130	0
		10,000 to 19,999	330	490	560	580	1,000	240	200	0
		20,000 to 39,999	790	550	900	660	1,350	320	270	0
		40,000 and above	790	550	900	660	1,350	320	270	0
General cargo	DWT	up to 4,999	90	50	180	60	0	0	0	0
		5,000 to 9,999	240	130	490	180	110	110	100	0
		10,000 to 19,999	720	370	1,450	520	150	150	130	0
		20,000 and above	720	370	1,450	520	150	150	130	0
Liquefied gas tanker	m3	up to 49,999	240	240	360	240	1,000	200	200	100
		50,000 to 99,999	1,700	1,700	2,600	1,700	1,000	200	200	100
		100,000 to 199,999	2,500	2,000	2,300	2,650	1,500	300	300	150
		200,000 and above	6,750	7,200	7,200	6,750	3,000	600	600	300
Other liquids tanker	DWT	up to 999	500	500	750	500	1,000	200	200	100
		1,000 and above	500	500	750	500	1,000	200	200	100
Vehicle carrier	DWT	up to 9,999	800	500	1,100	500	310	300	250	0
		10,000 to 19,999	850	550	1,400	510	310	300	250	0
		20,000 and above	850	550	1,400	510	310	300	250	0
Ro-ro	DWT	up to 4,999	750	430	1,300	430	260	250	170	0

		5,000 to 9,999	1,100	680	2,100	680	260	250	170	0
		10,000 to 14,999	1,200	950	2,700	950	390	380	260	0
		15,000 and above	1,200	950	2,700	950	390	380	260	0
Refrigerated bulk	DWT	up to 1,999	520	570	560	570	270	270	270	0
		2,000 to 5,999	1,100	1,200	1,150	1,200	270	270	270	0
		6,000 to 9,999	1,500	1,650	1,600	1,650	270	270	270	0
		10,000 and above	2,850	3,100	3,000	3,100	270	270	270	0
Service, tug	GT	0 and above	100	80	210	80	0	0	0	0
Miscellaneous, fishing	GT	0 and above	200	200	200	200	0	0	0	0
Offshore	GT	0 and above	320	320	320	320	0	0	0	0
Service, other	GT	0 and above	220	220	220	220	0	0	0	0
Miscellaneous, other	GT	0 and above	150	150	430	410	110	110	90	0

Source: Table 17 of the Fourth IMO Greenhouse Gas Study (IMO, 2020, printed pp. 68 and 69). AE denotes the auxiliary engines and AB the auxiliary boilers. Size bands are inclusive at both ends and a ship is matched to the band containing its capacity. Liquefied gas tankers are indexed by cubic capacity, container ships by twenty-foot equivalent units, and the remaining types by deadweight or gross tonnage as shown. Removed for this table, but can be found in the original reference: Ferry, passenger only, Ferry, ro-pax, Cruise, Yacht.

Table C6. Small-engine overrides applied after the Table C5 lookup.

Installed MCR from (kW)	to (kW)	Auxiliary engine demand	Auxiliary boiler demand
0	150	zero	zero
150	500	5 per cent of installed MCR	from Table C5
500	and above	from Table C5	from Table C5

Source: Fourth IMO Greenhouse Gas Study (IMO, 2020, printed p. 68). Neither pilot vessel falls in the first two bands, but a fleet extension covering small coastal ships will.

Table C7. Baseline specific fuel consumption in grams per kilowatt hour, by engine type, fuel type and year of build.

Engine type	Fuel	Before 1983	1984 to 2000	2001 and later

Slow-speed diesel (SSD)	HFO	205	185	175
	MDO	190	175	165
	Methanol	not applicable	not applicable	350
Medium-speed diesel (MSD)	HFO	215	195	185
	MDO	200	185	175
	Methanol	not applicable	not applicable	370
High-speed diesel (HSD)	HFO	225	205	195
	MDO	210	190	185
LNG-Otto, dual fuel, medium speed	LNG	not applicable	173	156
LNG-Otto, dual fuel, slow speed	LNG	not applicable	not applicable	148 LNG plus 0.8 MDO pilot
LNG-Diesel, dual fuel	LNG	not applicable	not applicable	135 LNG plus 6.0 MDO pilot
LBSI	LNG	not applicable	156	156
Gas turbines	HFO	305	305	305
	MDO	300	300	300
	LNG	not applicable	not applicable	203
Steam turbines and boilers	HFO	340	340	340
	MDO	320	320	320
	LNG	285	285	285
Auxiliary engines	HFO	225	205	195
	MDO	210	190	185
	LNG	not applicable	173	156

Source: Table 19 of the Fourth IMO Greenhouse Gas Study (IMO, 2020, printed pp. 70 and 71). MDO refers to distillate marine fuels in general, including marine gas oil. The study converted values between fuels using assumed energy densities of 40,200 kJ per kg for HFO, 42,700 for MDO, 48,000 for LNG and 19,900 for methanol. Both pilot vessels are post-2001 slow-speed diesel builds, so this study selects the 175 and 165 g kWh^{-1} pair for the main engine, 195 and 185 for the auxiliary engines, and 340 and 320 for the boilers. Equation (10) load-corrects the main engine only. The study states that gas and steam turbines are not load dependent and always take the baseline value.

Table C8. Low load adjustment factors, all pollutants.

Engine load	**PM**	**NO_x**	**SO_x***	**CO_2***	**CO**	**CH_4**	**N_2O**	**NMVOC**	**BC***
2 per cent or below	7.29	4.63	1.00	1.00	9.7	21.18	4.63	21.18	1.00
10 per cent	1.38	1.22	1.00	1.00	1.97	2.18	1.22	2.18	1.00
20 per cent	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

*Source: Table 20 of the Fourth IMO Greenhouse Gas Study (IMO, 2020, printed p. 74). * These pollutants vary directly as a function of fuel consumption, which is itself a function of engine load, so no adjustment is applied to them. CO_2 therefore carries a factor of 1.00 at every load and this study applies no low load adjustment. This table allows an extension of this method to other pollutants and shows how much more sensitive particulate matter and methane are to low-load operation than CO_2 is. The study reports no fuel consumption or main-engine emissions below 7 per cent of maximum continuous rating.*

Table C9. Fuel-based CO_2 emission factors and carbon contents, all fuels.

Fuel type	**Carbon content**	**EF (g CO_2 per g fuel)**
HFO	0.8493	3.114

LSHFO 1.0 per cent	0.8493	3.114
MDO	0.8744	3.206
LNG	0.7500	2.750
Methanol	0.3750	1.375

Source: Table 21 of the Fourth IMO Greenhouse Gas Study (IMO, 2020, printed p. 74), which takes the factors from the 2018 EEDI Guidelines. Low-sulphur HFO is assigned the same carbon content and emission factor as HFO, which is why the 2020 sulphur cap creates no branch in the CO_2 calculation. For engines burning a pilot fuel, the CO_2 from the pilot fuel is incorporated into the factor by weighting the fuel mix by CO_2 mass emitted.